

Electric Fish Tune Themselves to Musical Intervals

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1 Abstract

Some musical intervals are perceived as smooth and consonant, while others sound harsh and dissonant. In humans, this bias has long been attributed to the limited frequency resolution of the cochlea. However, many studies show that this explanation is insufficient and instead often suggest that cultural influences shape this preference.

Recently, we observed that weakly electric fish adjust the frequency of their electric organ discharges to form ratios corresponding to musical intervals. Since electroreceptors share many properties with inner hair cells, but process frequencies in a single channel, this system allows us to investigate the neural basis of consonance perception at the single cell level.

The current work aimed to reveal potential reasons for the bias towards musical intervals in weakly electric fish. Analyzing spectral differences between musical and 'mistuned' intervals revealed that mistuned intervals produce additional low-frequency amplitude modulations (AMs) that remain absent in musical intervals. These higher-order low-frequency AMs could facilitate the encoding of high difference frequencies in higher electrosensory brain areas.

Through a combination of computational modeling and behavioral experiments, we could show that these higher-order low-frequency AMs could be represented outside the electrosensory periphery and drive behavioral responses, indicating their ecological relevance.

Furthermore, we demonstrate that the non-linearities needed to generate such higher-order low-frequency AMs are biologically plausible and can be implemented at the single cell level. Hence, our findings suggest that the neural basis of consonance perception does not depend on complex auditory computations, but emerges from fundamental processing mechanisms of the nervous system.

2 Zusammenfassung

Bestimmte musikalische Intervalle werden generell als harmonisch oder *konsonant* empfunden, während andere als unangenehm oder *dissonant* beschrieben werden. Lange wurde diese unterschiedliche Wahrnehmung auf die limitierte Frequenzauflösung der Cochlea zurückgeführt. Jedoch konnte inzwischen gezeigt werden, dass diese Erklärung unzureichend ist.

Wir konnten beobachten, dass Paare von schwach-elektrischen Fischen die Frequenzen ihrer Felder so anpassten, dass sich Verhältnisse ergaben, die denen von musikalischen Intervallen entsprachen. Da das elektrosensorische System im Gegensatz zum auditorischen System nur einen einzelnen Frequenzkanal nutzt, erlaubt dieser Modellorganismus uns, die neuronalen Grundlagen der Konsonanzwahrnehmung auf Einzelzellebene zu untersuchen.

Unsere Analysen zeigten, dass 'verstimmte' Intervalle Amplitudenmodulationen (AMs) mit niedrigen Frequenzen erzeugen, die bei musikalischen Intervallen fehlen. Diese AMs könnten zur Enkodierung hoher Differenzfrequenzen genutzt werden. Mit Hilfe von Zellsimulationen konnte gezeigt werden, dass solche AMs in höheren Hirnarealen repräsentiert werden können. Zudem reagierten Fische auf verstimmte Intervalle mit einer Änderung ihrer Frequenz, was die ökologische Relevanz dieser niederfrequenten AMs unterstreicht.

Für die Entstehung dieser niederfrequenten AMs sind jedoch nichtlineare Signaltransformationen notwendig. Wir konnten zeigen, dass eine nichtlineare Operation, wie sie an den Elektrorezeptoren implementiert sein könnte, ausreicht, um diese AMs zu erzeugen. Damit weisen unsere Ergebnisse darauf hin, dass die neuronale Grundlage der Konsonanzwahrnehmung aus grundlegenden Mechanismen der Signalverarbeitung auf Einzelzellebene hervorgeht.

3 Introduction

Sounds are able to greatly influence our emotions. While birdsong, for example, is calming to most people, hearing a crying baby evokes feelings of distress. Similarly, the same visuals in a movie can be interpreted by the viewers in drastically different ways depending on the music the creators chose to pair with the scene. Composers play with the emotions evoked by sounds based on two fundamental concepts of music theory, namely consonance and dissonance. Consonance and dissonance perception can occur when two tones of the musical scale are played simultaneously or back-to-back, with some combinations generally being perceived as pleasant or consonant, while others are regarded as unharmonious and aversive, or dissonant (Malmberg 1918). Although consonance and dissonance have been fundamental concepts in music theory for a long time, it is still debated *why* we perceive certain combinations of tones as more pleasing than others (Helmholtz 2009; Plomp and Levelt 1965; Shapira Lots and Stone 2008). Furthermore, systematic research of this phenomenon has so far only concentrated on the differences between consonant and dissonant tone combinations, while dismissing the huge range of frequency combinations outside the musical system. To date, most explanations are either based on specific processing mechanisms of the mammalian inner ear or cultural influences, although none is able to explain this phenomenon in its entirety (see Di Stefano et al. 2022 for a review).

Recently, we observed weakly electric fish in their natural habitat to adjust the frequencies of their electric fields to correspond to musical intervals of Western tuning systems. This observation demonstrates that musical ratios play a role in sensory processing, independent of the specific constraints imposed by the mammalian auditory system. Instead, the fact that even electric fish tune themselves to musical intervals suggests that there is a fundamental, shared mechanism on a single cell level that is able to explain the differential percept of certain tone combinations.

3.1 Weakly Electric Fish as a Model Organism in Neuroscience

Weakly electric fish are an exotic but powerful model organism in neuroscience for studying general principles of sensory processing in natural contexts. Using a specialized electric organ, they produce low-amplitude discharges that generate an electric field around their bodies, the

so-called electric organ discharge (EOD; Engler and Zupanc 2001; Lissmann 1951; Stamper et al. 2012). Perturbations of this field caused by objects whose electric potential differs from the surrounding water can be sensed by the fish through specialized electroreceptors scattered on their body surface. These adaptations enable *active electroreception*, the ability to probe the environment using their self-generated electric field (Crampton 2019; Oboti et al. 2025; Von der Emde 1999).

Active electroreception has evolved independently in two teleost lineages, the African Mormyroidea and the South American Gymnotiformes (Bullock and Hopkins 2005; Crampton 2019). In each group, representatives of both pulse- and wave-type species can be found. Pulse-type fish, such as *Gnathonemus petersii*, irregularly produce discharges that can be separated by longer pauses. Wave-type species, in contrast, continuously emit a quasi-sinusoidal EOD with species- and individual specific frequencies of up to 2 kHz (Bullock and Hopkins 2005; Knudsen 1975; Salazar et al. 2013). Well-known examples of wave-type species include *Apteronotus leptorhynchus* and *Eigenmannia virescens*, which both occur naturally in lowland river basins throughout Central and South America (Crampton 2019; Oboti et al. 2025).

The ability to actively generate an electric field that is used to perceive the environment makes weakly electric fish an excellent model organism for neuroethological research. Recent advances in hardware development allow recordings of electric fields in natural habitats with minimal interference by the experimenter. Custom-built electrode arrays arranged in a 2D grid are commonly used for this purpose. As fish exhibit an individual EOD frequency that stays stable over longer time periods, such setups enable tracking of individuals across time and space while capturing natural behavior and social interactions (Henninger et al. 2018; Migliaro et al. 2025; Raab et al. 2022). Furthermore, electrophysiological paradigms in weakly electric fish have been well established during the past decades. Mimicking a conspecific’s electric signals while performing electrophysiological recordings allows researchers to assess the mechanisms of sensory processing from the receptor level to higher electrosensory areas (Bastian 1974; Fleishman 1992; Stöckl et al. 2014). This unique combination of field observations coupled with controlled laboratory experiments provides a powerful framework to investigate the neural system not in isolation, but as part of an organism interacting with the environment.

3.1.1 Electric Fields in Social Contexts

Fish do not use their electric fields solely to navigate the environment and locate small objects in their surroundings. Especially in wave-type species, interactions between the fields of two or more individuals play an important role in social behavior (Henninger et al. 2018; Walz et al. 2014; Zakon and Smith 2002). When the fields of two nearby fish superimpose, they give rise to amplitude modulations (AMs), also referred to as *beats*, with frequencies equal to the difference between the two EOD frequencies (Heiligenberg 1991; Stamper et al. 2010). Beats can convey social information, for example the sex of the nearby conspecific. This is because wave-type fish generally exhibit a sexual dimorphism in their EOD frequencies, with females tending to have higher frequencies than males (Hagedorn and Heiligenberg 1985; Henninger et al. 2018). As a result, opposite-sex interactions typically produce high-frequency beats, whereas same-sex interactions often generate low-frequency beats.

Low-frequency beats can elicit a behavioral response in most wave-type species, the so called Jamming Avoidance Response (JAR). The JAR manifests as a gradual shift in EOD frequency over the time course of multiple seconds, usually away from the EOD frequency of the close-by conspecific (Watanabe and Takeda 1963). For example, *Eigenmannia* will increase its frequency when presented with a lower-frequency signal, and decreases it when the stimulus has a higher frequency. Other genera, such as *Sternarchus*, exclusively produce upwards shifts of EOD frequency (Bullock et al. 1972). It is assumed that fish aim to increase the difference frequency to avoid sensory interference, as low-frequency beats may impair the fish' ability to electrolocate (Heiligenberg et al. 1978; Weimar et al. 2025). Although this behavior has been described extensively under laboratory conditions, a clear demonstration of the JAR in the fish' natural habitat remains lacking.

However, other types of EOD frequency modulations have been observed frequently in both natural and laboratory settings (Henninger et al. 2018; Zupanc et al. 2006). *Chirps* are brief, transient upward shifts of EOD frequency that have been associated with social behaviors such as courtship and aggression (Hagedorn and Heiligenberg 1985; Zakon et al. 2002). Their duration ranges from few tens to several hundreds of milliseconds, and the magnitude of frequency change can reach several hundred Hertz. Based on these parameters, chirps can be classified into four different types. One commonly observed type are *small chirps*, which have short

durations and small frequency increases and are produced by both males and females. *Long chirps*, in contrast, are characterized by large frequency excursions and extended durations, and occur predominantly in females (Bastian et al. 2001; Engler and Zupanc 2001; Henninger et al. 2018). Because chirps are produced primarily in social contexts, they are thought to serve communication purposes, with each type of chirp potentially conveying distinct information (Hagedorn and Heiligenberg 1985; Hopkins 1974; Smith 2013). However, this interpretation has been challenged recently (Oboti et al. 2025).

3.1.2 Neural Processing of Electrosensory Signals

In order to sense perturbations of their electric field, weakly electric fish possess two types of specialized tuberous electroreceptors that are distributed across their body surface. The first type of electroreceptors, the so-called T-units ('time coders') fire one spike per EOD cycle at a fixed phase. As such, T-units provide precise temporal information in relation to the zero-crossing of the EOD waveform (Kashimori et al. 1996; Nelson et al. 1997). In contrast, P-units ('probability coders') encode information about AMs of the EOD (Bastian 1981a; Nelson et al. 1997). Their firing probability scales with stimulus intensity. Hence, in the absence of external electric fields, P-units fire irregularly during the upward phase of the fish's own EOD (Gussin et al. 2007; Kashimori et al. 1996). If the EOD is amplitude modulated, however, the P-units' spiking probability will decrease at the low-amplitude phases of the AM and increase during its high-amplitude phases (Bastian 1981a; Kashimori et al. 1996; Nelson et al. 1997).

P-unit tuning has been extensively investigated for beat frequencies up to 300 Hz in the past (Bastian 1981a; Benda et al. 2006; Walz et al. 2014). Though, especially during opposite-sex and inter-species interactions, the frequency difference between fish can exceed this range by hundreds of Hertz (Henninger et al. 2018; Henninger et al. 2020). For this reason, Barayeu et al. (2023) recently investigated if and how P-unit receptors are able to represent very large difference frequencies. By performing electrophysiological recordings within the anterior lateral line nerve, which contains the afferent fibers from electroreceptors (Carr et al. 1982), they demonstrated that P-units respond to difference frequencies exceeding 2000 Hz. However, the observed responses did not reflect a faithful tracking of the predicted beat frequency. Instead, P-units profit from the aliasing structure imposed by the fish's own EOD when encoding such high difference frequencies. As the EOD effectively acts as a sampling carrier for the AM, beat

frequencies above half of the EOD frequency fold back around integer multiples of the EOD frequency, as predicted by the Nyquist theorem. Consequently, very high beat frequencies reappear as lower-frequency aliased components that fall within the P-unit's tuning range.

Furthermore, Barayeu et al. (2023) evaluated potential mechanisms through which these aliased components are extracted from the original signal. They demonstrated that non-linearities commonly used to retrieve AM frequencies, such as squaring or the Hilbert transform (Middleton et al. 2007; Stamper et al. 2012), could not account for the experimentally measured P-unit responses. Instead, a cubed threshold operation was sufficient to extract aliased components below half the EOD frequency. This non-linearity is reminiscent of the transformation introduced at the ribbon synapse of auditory inner hair cells (Peterson and Heil 2019). As electroreceptors also connect to their afferents via ribbon synapses (Szabo 1965; Wachtel and Bruce Szamier 1966), it is plausible that a similar synaptic transformation implements this cubed threshold operation in the electrosensory system.

Downstream of the ribbon synapse, the afferents of electroreceptors trifurcate and project onto pyramidal cells in the electrosensory lateral line lobe (ELL). Thereby, each branch innervates one of the three parallel topographic maps of this brain area, which comprise the centromedial segment (CMS), the centrolateral segment (CLS), and the lateral segment (LS; Carr et al. 1982; Heiligenberg and Dye 1982; Márquez et al. 2013). These topographic maps differ in their receptive field sizes (Krahe et al. 2008; Maler 2009), but each contains two distinguishable classes of pyramidal cells. *E-cells*, which increase their firing rate in response to amplitude increases, and *I-cells*, that reduce their activity when the amplitude increases (Bastian 1981a; Maler 1979; Márquez et al. 2013). Pyramidal cell outputs are then relayed to midbrain and thalamic regions, where responses become increasingly selective to specific stimulus features (Vonderschen and Chacron 2011; Wallach et al. 2022).

3.1.3 Representing Fast Beats Beyond Electroreceptors

Although much of the neuronal circuitry underlying electrosensory processing is well researched, some aspects are still not fully understood. For example, it remains unclear how large difference frequencies are represented beyond the electrosensory periphery. Pyramidal cells have rarely been observed to fire at frequencies above 50 Hz, with few exceptions (Bullock 1969; Chacron 2006). Even though P-units can follow beats faithfully well beyond this range (Barayeu et

al. 2023), higher sensory areas most likely depend on an alternative way of representing fast beat frequencies. Recent observations of electric fish behavior in their natural habitat may provide a potential clue to this puzzle. Exploratory analyses revealed that fish adjust their EOD frequencies to form simple integer ratios (Figure 1). Remarkably, these ratios correspond to the frequency ratios of musical intervals. As these simple frequency ratios are known to create characteristic spectral patterns, they could be exploited by the electrosensory system in order to represent fast beat frequencies without following the high firing rates of P-units.

3.2 Foundations of Musical Intervals and Consonance

When two sounds from a musical scale are played simultaneously or in succession, their combination is either perceived as harmonious and pleasant or harsh and aversive. These combinations of sounds are referred to as *musical intervals*, and the differential percept they elicit has long intrigued musicians and scientist alike (Di Stefano et al. 2022; Helmholtz 2009; Malmberg 1918). Early accounts of this phenomenon can be dated back to ancient Greece. Pythagoras, for example, observed that the pleasantness of sounds generated by plucking two strings depends on the simplicity of the ratio between their lengths (Bowling and Purves 2015; Crocker 1963). Much later, Helmholtz published his influential work *On the Sensations of Tone*, in which he systematically investigated the differences between pleasant *consonances* and unpleasant *dissonances* (Di Stefano et al. 2022; Helmholtz 2009). Although numerous attempts have been made to describe why some intervals sound consonant while others do not, to date no single explanation fully accounts for this phenomenon in its entirety.

3.2.1 Formalizing Intervals and Their Spectral Properties

A musical interval is defined as the difference in pitch between two notes on a musical scale (Prout 2011). This difference can be expressed as the frequency ratio between the fundamental frequencies f_1 and f_0

$$\frac{f_1}{f_0} = \frac{m}{n} \tag{1}$$

for integers m, n (Helmholtz 2009). Table 1 lists several commonly used intervals in Western music. As already observed by Pythagoras, intervals that are perceived as pleasant typically correspond to those with simple integer ratios, such as the octave, perfect fourth, and perfect

fifth (Crocker 1963; Di Stefano et al. 2022). Indeed, such intervals exhibit characteristic spectral traits which are thought to contribute to their perceptual pleasantness. Naturally occurring sounds are not pure sine waves, but contain harmonics at integer multiples of their fundamental frequency with gradually decreasing amplitude (Moore 2012). For combined sounds with simple integer ratios, harmonics frequently coincide. For instance, in an octave with $f_1/f_0 = 2 : 1$, every harmonic of f_1 aligns with every second harmonic of f_0 . Similarly, in a perfect fourth with $f_1/f_0 = 4 : 3$, every third harmonic of f_1 overlaps with every fourth harmonic of f_0 . Hence, as the ratio becomes more complex, fewer harmonics coincide (Helmholtz 2009; Plomp and Levelt 1965).

When combined sound waves are transformed by thorough a non-linearity, non-overlapping harmonics interact to give rise to additional spectral components known as intermodulation (IM) products. IM products appear at

$$f_{IMP} = |mf_1 \pm nf_0| \quad (2)$$

for integers m, n . Hence, the frequency ratio directly influences the distribution of such components (Hemmi 2002; Wass 1948). Intervals with simple integer ratios generate comparatively few IM products, as many of their harmonics coincide. In contrast, more complex ratios produce a larger number of irregularly spaced IM products, because most harmonics do not align. The order of an intermodulation product is given by the sum $m + n$ and predicts the strength and frequencies at which they appear. Third-order IM products, for instance, usually appear close to the fundamental frequencies as well as in the low-frequency regime (Anritsu Corporation 2004).

Similarly, small deviations from exact integer ratios also affect the occurrence of IM products. For an interval mistuned by a factor ϵ , such that

$$\frac{f_1}{f_0} = \frac{m}{n} + \epsilon, \quad (3)$$

previously coinciding harmonics are shifted away from one another. The smaller the mistuning factor ϵ , the smaller the frequency difference between such harmonics. Their interactions therefore also introduce IM products in the low-frequency regime (Trulla et al. 2018). In the temporal

domain, such low-frequency IM products manifest as higher-order low-frequency (HOLF) AMs that are superimposed on the first-order beat.

Table 1: Common musical intervals. Overview of musical intervals commonly used in Western music, their respective abbreviations, and frequency ratios as integer ratios and decimal numbers.

Name	Abbreviation	$f_1 : f_0$	f_1/f_0
Unison	P1	1:1	1.0
Minor Second	m2	16:15	$1.0\bar{6}$
Major Second	M2	9:8	1.125
Minor Third	m3	6:5	1.2
Major Third	M3	5:4	1.25
Perfect Fourth	P4	4:3	$1.\bar{3}$
Tritone	TT	7:5	1.4
Perfect Fifth	P5	3:2	1.5
Minor Sixth	m6	8:5	1.6
Major Sixth	M6	5:3	$1.\bar{6}$
Subminor Seventh	sm7	7:4	1.75
Minor Seventh	m7	9:5	1.8
Major Seventh	M7	15:8	1.875
Octave	P8	2:1	2.0

3.2.2 Competing Theories of Consonance Perception

Interactions between non-overlapping harmonics have long provided the basis for influential accounts of consonance and dissonance perception. Helmholtz, for instance, attributed the aversive quality of dissonances to the phenomenon of *roughness* (Di Stefano et al. 2022; Helmholtz 2009). Roughness thereby corresponds to AMs in the range of roughly 20-300 Hz (Di Stefano et al. 2022; Fastl and Zwicker 2006; Parncutt 1989). These specific modulations are proposed to occur when individual frequency components fall within the same critical band of the cochlea, causing overlapping regions of the basilar membrane to vibrate simultaneously (Helmholtz 2009; Plomp and Levelt 1965). Consonant intervals produce less roughness and are thus perceived as more pleasant. Several findings, however, suggest that roughness alone cannot fully account for consonance perception. For example, it has been found that dissonant intervals are still perceived as aversive even when their amplitude modulations exceed the proposed roughness range (Bidelman 2013; Bidelman and Krishnan 2009; McDermott et al. 2010). Furthermore, studies of dichotic listening, in which each tone of the interval is presented to one ear sep-

arately, thus preventing frequency interactions on the cochlea, have produced mixed results. Some report decreased consonance rankings (Bones et al. 2014), whereas others suggest that consonance rankings remain unchanged (Bidelman and Krishnan 2009; Houtsma and Goldstein 1972; McDermott et al. 2010). Furthermore, individuals with congenital amusia exhibit a aversion to beating comparable to controls, but lack a preference for consonances over dissonances (Cousineau et al. 2012).

More recently, the concept of *harmonicity* has been emphasized in the study of consonance and dissonance perception as an alternative explanation to roughness. Harmonicity is a measure expressing how similar a sound spectrum is to a harmonic series, that is, whether the sound exhibits a full set of integer multiples of its fundamental frequency (Di Stefano et al. 2022). In the temporal domain, harmonicity corresponds to periodicity, which predicts the repetitiveness of a waveform (Harrison and Pearce 2020). As harmonics align frequently in consonant intervals, they produce highly periodic waveforms, whereas dissonant interval exhibit a more irregular temporal structure. Behavioral and modelling studies could demonstrate that harmonicity predicts consonance (McDermott et al. 2010; Smit et al. 2019). Nevertheless, the preference for harmonicity also appears to be largely culturally shaped (Maher 1976; McDermott et al. 2016), challenging its interpretation as an universal explanation for the pleasant nature of consonance.

3.2.3 Neural Representation of Consonance and Dissonance

The limitations and uncertainties raised by purely psychophysical accounts of consonance and dissonance have inspired the investigation of the representation of musical intervals in the auditory system. Neurophysiological studies were able to reveal processing differences between consonant and dissonant intervals. For instance, Fishman et al. (2001) could show that dissonant intervals elicit strong phase-locking to low-frequency IM products in the auditory cortex of macaques. Musical intervals did not generate such a response pattern, thereby providing neural evidence for Helmholtz’ account of roughness.

Recordings from the auditory nerve in cats, on the other hand, have found that consonant intervals produce interspike interval distributions that reflect the harmonic structure of simple integer ratios. The strength of the fundamental frequency in the distribution could also predict consonance perception, linking neural activity to perceptual outcomes (Tramo et al. 2001).

These findings, in turn, support the idea that harmonicity plays a key role in shaping the perception of consonant intervals.

Thus, although neurophysiological studies were able to demonstrate clear differences in the neural representation of consonant and dissonant intervals at different levels of the auditory system, interpretations about the spectral and temporal features driving these differences remain ambiguous. However, given that most available neurophysiological evidence stems from non-human animals, it appears likely that the neural mechanisms underlying the differential percepts of musical intervals are evolutionarily conserved. Comparing these mechanisms across species may therefore provide valuable insights into how the auditory system processes complex sounds, which may eventually yield a better understanding of consonance and dissonance perception.

3.3 Objectives and Hypotheses

Despite extensive research over multiple centuries, the mechanisms underlying consonance and dissonance perception remain unresolved. Although culture and exposure most likely influences explicit preferences (McDermott et al. 2016; Smit et al. 2019), neural representations of consonant and dissonant intervals differ across species (Fishman et al. 2001; Tramo et al. 2001). This suggests that fundamental and evolutionarily conserved neural processes contribute to this phenomenon. The findings presented in the current study support this view.

Recently, we observed that weakly electric fish tune their EOD frequencies to integer ratios corresponding to musical intervals (Figure 1). Because the electrosensory system processes frequencies within a single channel, we hypothesize that the spectral properties unique to musical intervals are generated at fundamental stages of neuronal processing. Specifically, we suggest that the single non-linearity implemented at the level of individual electroreceptors, which is needed to retrieve first-order beat frequencies (Barayeu et al. 2023), is sufficient to generate spectral differences between musical and non-musical intervals. Furthermore, we propose that 'mistuned' intervals generate additional low-frequency amplitude modulations through this mechanism. These higher-order low-frequency (HOLF) beats potentially facilitate the encoding of large difference frequencies in higher electrosensory areas.

To evaluate these predictions, we first examined whether the cubed threshold non-linearity of electroreceptors can generate higher-order amplitude modulations that distinguish musical from non-musical intervals. We then investigated how these additional amplitude modulations, specifically HOLF beats at mistuned intervals, are represented in the electrosensory system using a population of simulated electroreceptors. Finally, we tested whether HOLF beats carry a behavioral relevance in a classical jamming avoidance paradigm.

By integrating theoretical considerations, computational modeling, and behavioral experiments, the current work aims to identify a simple and biologically plausible mechanism that produces differential responses to musical and mistuned intervals. Not only will this study provide insights into how fast beat frequencies may be represented beyond the electrosensory periphery, but it also offers a novel perspective on the phenomenon of consonance and dissonance perception.

4 Materials and Methods

4.1 Field Data

4.1.1 Data Acquisition

To investigate whether and under which conditions pairs of *Eigenmannia* assume frequency ratios corresponding to musical intervals (Table 1), three previously collected field datasets were analyzed (Table 2). All recordings were obtained in the natural habitats of weakly electric fish using custom-built multi-electrode arrays. These arrays enable continuous recordings of electric organ discharges (EODs) over multiple days with minimal disturbance to the animals (Henninger et al. 2018; Migliaro et al. 2025; Raab et al. 2022). Hence, these datasets capture natural behavior and social interactions between fish. Because the EOD frequency of wave-type species is largely stable over minutes up to hours, individuals within these datasets may be identified by their fundamental frequency and tracked over time and space (Raab et al. 2022).

Dataset 1 (D1) was obtained in the Darién region of Panama at the onset of rainy season in March 2012, which coincides with the breeding season of several species of weakly electric fish, including *Eigenmannia virescens* (Hagedorn and Heiligenberg 1985; Hopkins 1972). These recordings have been published previously (Henninger et al. 2018), and a preprocessed version is available at <https://doi.org/10.12751/g-node.87584d>. Data were acquired using a 9×6 electrode grid (240×150 cm) controlled through the custom written *Fishgrid* software (<https://github.com/bendalab/fishgrid>) at a sampling rate of 20 kHz. D1 constitutes the smallest of the utilized datasets, containing approximately 30 hours of EOD recordings.

Dataset 2 (D2) was collected in the Río Cañita, Panama, during the rainy season in May 2014. A 12×5 electrode grid (440×160 cm) was used and recordings were controlled with the same software as in D1. Data were sampled at a rate of 20 kHz, and a total of almost 89 hours of EOD recordings have been collected.

Dataset 3 (D3) was acquired in FLONA do Amapá, Brazil, between late September and the beginning of October 2023. In contrast to the first two datasets, these recordings were obtained outside the local rainy season. Signals were captured using electrode lines with 16 channels spaced 25 cm apart at a sampling rate of 96 kHz. Recordings were controlled using the custom

Table 2: Datasets used in Fielddata Analysis. Overview of location, season, hardware configuration, and total duration of recordings of the three field datasets used to examine frequency ratios assumed by *Eigenmannia*.

Dataset #	Location	Date	Electrodes	Duration
1	Darien, Panama	March 2012 (rainy season)	9×6	30h 13min
2	Río Cañita, Panama	May 2014 (rainy season)	12×5	88h 58min
3	Río Falsino, Amapá, Brazil	September–October 2023 (dry season)	1×16	199h 36min

written TeeGrid software (<https://github.com/janscience/TeeGrid>). This dataset is the largest of the three, comprising roughly 199 hours of EOD recordings collected across several sites in the Río Falsino.

4.1.2 Data Analysis

The data recorded using multi-electrode arrays are complex both in dimensionality and signal structure, as multiple fish are recorded simultaneously. Extracting the EOD signals of individual fish therefore requires dedicated preprocessing. Signal extraction and all other subsequent analyses were performed using Python 3.12.3 and the *thunderlab* (<https://github.com/bendalab/thunderlab>) and *thunderfish* (<https://github.com/bendalab/thunderfish>) packages, developed by the Neuroethology Lab at the University of Tübingen for handling electrode grid data.

For each electrode channel, power spectra were computed in 10-second windows using Welch’s method with 50% overlap. The fundamental frequencies of individual fish were identified using the *harmonic_groups* algorithm in *thunderfish*. This algorithm groups peaks detected in the power spectra according to the known harmonic structure of EOD signals for different species and returns the fundamental frequency of all individuals detected in each electrode channel and 10-second window. Four faulty channels in D3, for which power spectra could not be computed due to a lack of data, were excluded from this analysis. Extracted fundamental frequencies were saved as .csv files along with an index indicating both electrode channel and time window of detection as well as the corresponding signal power in decibel, computed as

$$P_{dB} = 10 \log_{10}(P). \quad (4)$$

Note that the power P was not normalized to a reference power, such that the maximum power in each power spectrum is set to 0 dB. Because the current study focuses on musical intervals in *Eigenmannia*, only signals with a fundamental frequency within this genus' characteristic frequency range (240 – 540 Hz; Henninger et al. 2020) were retained for further analysis.

In each non-overlapping 10-second time window, the two strongest EOD signals (highest power in dB) at each electrode were selected, approximately corresponding to the two fish closest to the given electrode. Note that it cannot be excluded that another fish, albeit with a weaker EOD, was closer to the channel. Time windows and channels where fewer than two fish were detected in the signal were excluded from this analysis. The EOD frequency ratio between the two closest fish was then computed as $r_{fp} = f_1/f_0$, with f_1 being defined as the higher frequency. Results were stored in lists for further computations.

Histograms of the frequency ratios assumed by pairs of fish were generated using *NumPy*'s *histogram* function with 2000 logarithmically spaced bins between ratios 1 and 2 (i.e., one octave). To quantify how accurately and precisely pairs of fish assumed frequency ratios corresponding to musical intervals, deviations within a $\pm 1\%$ tolerance window around each musical interval were analyzed. For each musical interval with ratio r_{mi} , all fish pairs at all time points whose frequency ratio r_{fp} fell within $(1 - 0.01)r_{mi} < r_{fp} < (1 + 0.01)r_{mi}$ were extracted. Since this tolerance window scales with the musical interval itself, its width is smaller for intervals near the unison (e.g., 0.99 – 1.01 for $r_{mi} = 1$) than for intervals closer to the octave (e.g., 1.485 – 1.515 for $r_{mi} = 1.5$). For each musical interval, the deviation of each fish pair from r_{mi} was $\Delta r = r_{fp} - r_{mi}$. The mean deviation $\overline{\Delta r}$ thus reflects the accuracy with which fish assumed musical interval ratios. Values near zero indicate that fish generally tended to assume ratios very close to r_{mi} , while negative and positive values imply that fish often assumed slightly lower or higher ratios, respectively.

The standard deviation σ_r , on the other hand, quantifies the precision of fish matching musical intervals. Small values indicate a tight clustering of fish pair ratios around $\overline{\Delta r}$, while larger values indicate a more broad distribution within the tolerance window. Since the widths of the tolerance windows differ across musical intervals, the raw σ_r is not comparable between intervals. Hence, we additionally normalized it by the standard deviation of a uniform distribution of equal width, $\sigma_{uniform} = \frac{b-a}{\sqrt{12}}$, where a and b denote the boundaries of the interval. The normalized

standard deviation was computed as $\sigma_{norm} = \frac{\sigma_r}{\sigma_{uniform}}$, yielding a unitless measure where values < 1 indicate that clustering was tighter than for a normal distribution (high precision), and values > 1 suggest a broader scattering of fish assumed ratios (low precision).

4.2 Spectral Analysis

Consonant and dissonant musical intervals are known to differ in their spectral structure (Helmholtz 2009; Plomp and Levelt 1965). To investigate how musical intervals differ from their 'mistuned' variants in weakly electric fish, waveforms corresponding to the EOD of *Eigenmannia virescens* were simulated using amplitude and phase parameters as provided by the *fakefish* module from the *thunderfish* Python package.

For visual inspection of differences between musical and mistuned intervals, a receiver fish with a fundamental frequency $f_0 = 100$ Hz and signal amplitude of 1 was simulated, yielding an unitless waveform. The signal of a sender fish was added to the receiver's signal with a relative signal strength of 80% (contrast). The sender fish's frequency f_1 was varied systematically across the range of $0.9f_0$ to $2.1f_0$ in steps of $0.001f_0$. Frequencies corresponding to musical intervals with irrational frequency ratios, such as the perfect fourth with $f_1/f_0 = 1.\bar{3}$, were added manually. Based on the prevalence of certain musical intervals observed in the field data (Figure 1), the analysis focused primarily on the major third ($f_1/f_0 = 1.25$), perfect fourth ($f_1/f_0 = 1.\bar{3}$), and perfect fifth ($f_1/f_0 = 1.5$). Although the major sixth ($f_1/f_0 = 1.\bar{6}$) occurred more often than the perfect fifth in D1 and D2, this interval was omitted from detailed analysis as it is symmetrical to the perfect fourth and thus is expected to exhibit a similar spectral structure.

As mistuning a musical interval based on a fixed difference in Hertz will lead to different degrees of mistuning depending on the magnitude of f_0 , mistuned intervals will be described by a mistuning factor ϵ , according to Equation 3.

The waveforms of different orders of beats were retrieved using custom written Python functions. Higher-order low-frequency (HOLF) beats were computed from rectified versions of the combined signals. First, peaks on this rectified signal were detected using *SciPy*'s *find_peaks* function with a prominence threshold at 90% of the maximum signal amplitude. To obtain a continuous waveform, cubic spline interpolation (*scipy.interpolate.CubicSpline*, boundary con-

dition = "natural") was applied between detected peaks. First-order beat waveforms were computed similarly, but with a prominence threshold of 10% of the maximum signal amplitude and using *scipy.interpolate.make_interp_spline*. The resulting signal was low-pass filtered using a 4th-order Butterworth filter with a cutoff frequency of 60 Hz. To align the beat AM with the HOLF AM, the low-pass filtered signal was subsequently multiplied by the HOLF waveform and divided by its mean.

In order to generate additional spectral peaks corresponding to the first-order beat frequency and other intermodulation products, a non-linear operation needs to be applied to the signal. Barayeu et al. (2023) found that thresholding the signal and raising it to a power of three is sufficient to explain the response properties of certain electroreceptors, so-called P-units. Power spectra were estimated from thresholded and cubed signals (sampling rate = 10 kHz) using Welch’s method with $nperseg = 2^{18}$, resulting in a frequency resolution of approximately 0.038 Hz. Peaks in the dB spectra were detected using *scipy.find_peaks* with a prominence threshold of 36 dB. This threshold was chosen by visually verifying the legitimacy of detected peaks at different prominence thresholds. Peaks corresponding to the beat frequencies and their harmonics were identified as the peaks closest to $f_{beat} = f_1 - f_0$ and integer multiples thereof with a ± 1 Hz tolerance. Peaks not corresponding to either the fundamentals, the beat frequency, or their respective harmonics were classified as intermodulation (IM) products, which are generated at frequencies according to Equation 2.

To quantify spectral differences for all ratios across the octave, simulations were repeated with $f_0 = 1000$ Hz and accordingly adapted f_1 . All peaks exceeding a prominence threshold of 10^{-5} were counted in the power spectrum of the thresholded and cubed combined signals. The number of peaks was plotted as a function of consonant and dissonant intervals to explore the potential relationship to perceived consonance. Pleasantness ratings for musical intervals were obtained from Bowling et al. (2018), McDermott et al. (2010), Shapira Lots and Stone (2008), and Trulla et al. (2018).

To further assess the presence and strength of HOLF beats across different frequency ratios, thresholded and cubed combined signals with $f_0 = 1000$ Hz were convolved with Gaussian kernels of two different widths. Convolution with a Gaussian kernel effectively acts as a low-pass filter, with the kernel width σ determining the cutoff frequency $f_c = \frac{1}{2\pi\sigma}$. Hence, for the narrow

kernel with $\sigma = 0.5$ ms, $f_c \approx 318$ Hz, while for the broad kernel with $\sigma = 5$ ms, $f_c \approx 32$ Hz. The convolution with the narrow kernel therefore preserves frequencies up to a few hundred Hertz, while the convolution with the broad kernel only retains frequency components in the order of a few tens of Hertz. To obtain a measure for the strength of such high- and low-frequency components present in the signal, the standard deviation σ_{sig} of the convolved output was calculated.

4.3 Model Cell Responses

To examine whether neurons in the electrosensory system are able to encode HOLF beats that arise at mistuned intervals, computational models of P-units provided by the Neuroethology Lab at the University of Tübingen were utilized (<https://github.com/bendalab/punitmodel>). This population consists of 39 simulated P-units implemented as leaky integrate-and-fire (LIF) neurons with adaptation and intrinsic noise. Model parameters for individual cells were fitted to experimental recordings in *Apteronotus leptorhynchus*. The model processes input stimuli by applying a dynamic threshold followed by a low-pass filter. The threshold is modulated by an adaptation current, reducing the spike probability for identical inputs. Spikes are generated whenever the membrane potential crosses the threshold, and integration of the membrane potential is temporarily paused after each spike to simulate the refractory period of neurons. The intrinsic noise included in the model introduces trial-to-trial variability, thereby mimicking the behavior of real electroreceptors.

4.3.1 Stimulus Generation and Simulations

Stimuli were generated from simulated *Eigenmannia* waveforms of different frequencies, combined to produce a stimulus set covering a wide range of frequency ratios. Using harmonic waveforms was essential in this context to ensure that IM products would arise in the signal, as they do for real EODs. However, since the model was originally fitted to respond to pure sinusoidal waveforms and combinations thereof, the appropriate amplitude of *Eigenmannia* waveforms that results in comparable firing rates needed to be determined first. A bisection method yielded a scaling factor of 0.96 relative to stimuli without harmonics.

Stimuli had a duration of 20 seconds sampled at 20 kHz. Receiver fish waveforms were simulated with a frequency f_0 that matched the EOD frequency of the fish from which the receptor was recorded. Receiver waveforms were scaled by the previously determined factor of 0.96. Sender waveforms were generated with frequencies ranging from $0.9f_0$ to $2.1f_0$ in steps of $0.001f_0$ and a amplitude of 80% relative to the receiver waveform. Sender frequencies corresponding to musical intervals with irrational frequency ratios were added manually. The two generated waveforms were combined by addition in order to generate the final stimuli, which were presented to each cell for 150 trials. The model returned an array of spike times relative to the stimulus time vector for each trial.

4.3.2 Data Analysis

Spike times were converted into binary spike trains by assigning a value of 1 to each time bin in which a spike occurred, and 0 otherwise. Binary spike trains were then averaged across trials for each stimulus and cell.

Power spectra of the cell responses were estimated from the averaged binary spike trains using Welch’s method with $n_{\text{perseg}} = 2^{16}$ and converted to dB according to Equation 4. With the employed sampling rate of 20 kHz, this yielded a frequency resolution of approximately 0.076 Hz. Peaks in the dB spectra were detected using *SciPy*’s *find_peaks* with a prominence threshold of -50 dB. Peaks corresponding to the beat frequency $f_{\text{beat}} = f_1 - f_0$ and harmonics thereof were identified with a ± 1 Hz tolerance. All remaining peaks were classified as IM products.

Convolved firing rates were computed by convolving the averaged binary spike train with Gaussian kernels of different standard deviations σ . The narrow kernel ($\sigma = 0.5$ ms) approximates integration time windows of P-units, and the broader kernel ($\sigma = 5$ ms) was chosen to be within the range of experimentally determined pyramidal cells integration times (Berman and Maler 1998).

The strength of firing rate modulation for each stimulus and convolution kernel was quantified as the standard deviation σ_{FR} of the convolved firing rate. For visualizations, σ_{FR} values for each ratio were averaged across model cells.

4.4 Behavioral Experiments

4.4.1 Animals

The Jamming Avoidance Response (JAR) experiments were conducted on 11 individuals of the species *Eigenmannia virescens*. Since sex of the fish cannot be reliably determined by morphological characteristics in this species, the number of female and male fish is unknown. The individuals' EOD frequencies ranged from 318 to 446 Hz (362 ± 39 Hz SD, temperature corrected to 26°C). Temperature correction was done by fitting a linear regression model over the measured frequency – temperature points ($n = 4$ per fish) and predicting the EOD frequency at 26.0°C for each fish, resulting in a mean slope of 15.7 Hz/°C across individuals. Fish were housed either in solitude or in groups of up to five in tanks measuring 49 x 80 x 49 cm³ with a 12h day-night cycle. Due to the length of the stimulus protocol, each fish was tested on two separate days, with a one-day break between sessions. Upon transfer of the animals to the experimental tank, individuals were given five minutes of free exploration time to habituate to the new environment. In case the animal appeared stressed after this period, an additional five-minute acclimation period was provided.

4.4.2 Experimental Setup and Stimuli

Experiments were conducted in a tank measuring $50 \times 40 \times 40$ cm³, filled with 12 cm of water taken from the housing facility. A heating mat placed underneath the tank ensured that the temperature was maintained between 24°C – 26°C, comparable to housing conditions, minimizing temperature-dependent effects on EOD frequency. A PVC tube whose sides were replaced by a rough mesh ('chirp chamber') was positioned in the middle of the tank to offer shelter for the fish and minimize movement artifacts in the recordings. The EOD was continuously recorded using graphite electrodes (350×8 mm) positioned 2 cm anterior and posterior to the fish in the chirp chamber. Signals were amplified, with the gain adjusted to each fish' EOD amplitude to prevent clipping. Subsequently, signals were band-pass filtered (3 Hz - 1.7 kHz, DPA2-F, npf electronics, Tamm, Germany), and digitized at 40 kHz (PCI-6229, National Instruments, Austin TX, USA).

Attenuated stimuli (ATN-01M, npf electronics, Tamm, Germany) were delivered by another set of graphite electrodes (110×8 mm) placed perpendicular to the chirp chamber. Cal-

ibration of the electrodes, recording, and stimulus delivery were controlled using RELACS (<https://github.com/relacs/relacs.git>), and all data including metadata were stored in the .nix format (<https://github.com/G-Node/nix.git>) for reproducible analysis. The stimulus set consisted of sinusoidal signals simulating conspecifics at frequencies f_{stim} corresponding to four musical intervals, namely the unison, major third, perfect fourth, and perfect fifth (refer to Table 1 for a detailed description of frequency ratios), as well as eight mistuned versions of each (Equation 3; $\epsilon \pm 0.05, \pm 0.01, \pm 0.005, \pm 0.001$). The ratios furthest apart from the musical interval, that is, those with $\epsilon = \pm 0.05$, were intentionally placed approximately centered between two consecutive intervals, serving as control condition to evaluate whether behavioral responses were indeed specific to mistuned intervals rather than any arbitrary frequency ratio. Furthermore, the unison condition, which corresponds to classical JAR stimuli, verified that fish display a JAR at all. The other musical intervals were chosen based on their prevalence in the field data (Figure 1). Each frequency ratio was intended to be presented five times in randomized order. However, due to experimental inconsistencies, some frequency ratios were only presented four times to individual fish, whereas others were presented six times.

The strength of the stimuli was set to 80% (low contrast) or 300% (high contrast) of the fish’s initial EOD amplitude. Each trial, the stimulus was presented for 20 seconds, with 20 seconds before and 20 seconds after each stimulus, ensuring the fish could return to its baseline EOD frequency before the onset of the next trial.

4.4.3 Data Analysis

The recorded EOD voltage traces, corresponding time stamps, and experimental metadata were extraced from the saved .nix files using *relacsed_nix* (https://github.com/relacs/relacsed_nix). Instantaneous EOD frequency values were obtained using the *eod_times* function from the Benda Lab P-unit model repository (<https://github.com/bendalab/punitmodel>). This function estimates the times of positive zero crossings of a band-pass filtered version of the signal and computes the instantaneous EOD frequency as the inverse of the time interval between zero-crossings. The resulting EOD frequency trace was subsequently smoothed by convolving it with a moving average kernel with a length of 111 sample points of equal weight.

For each trial, the baseline EOD frequency f_{base} was computed as the mean EOD frequency for the last ten seconds before stimulus onset, ensuring that the fish has fully returned to its original frequency. Similarly, the EOD frequency during stimulus presentation f_{adj} was taken as the mean of the last ten seconds of stimulus presentation, such that it excluded most of the period of active frequency change. The relative frequency change in Hz was calculated as $\Delta EODf = f_{base} - f_{adj}$. Furthermore, frequency ratios before and after the fish's EOD frequency adjustment were computed as $f_1/f_{0_{bef}} = f_{stim}/f_{base}$ and $f_1/f_{0_{aft}} = f_{stim}/f_{adj}$, respectively. Averaged responses are reported as the median across all repeats of the same stimulus, either for individual fish or across all subjects.

5 Results

5.1 Fish Assume Musical Intervals in their Natural Habitat

To investigate how often *Eigenmannia* adjust their EOD frequencies to form ratios corresponding to musical intervals, data recorded in their natural habitat using multi-electrode arrays were analyzed. Datasets from different seasons and locations were included to assess potential environmental influences on this behavior. For each 10-second bin in the recordings, the ratio of the two strongest signals at each electrode was calculated, and the distribution (PDF) of all frequency ratios within one octave was computed (Figure 1).

In all datasets, peaks in the distribution of EOD frequency ratios were observed, although their position and prominence varied across seasons and recording sites. In Dataset 1 (D1), obtained

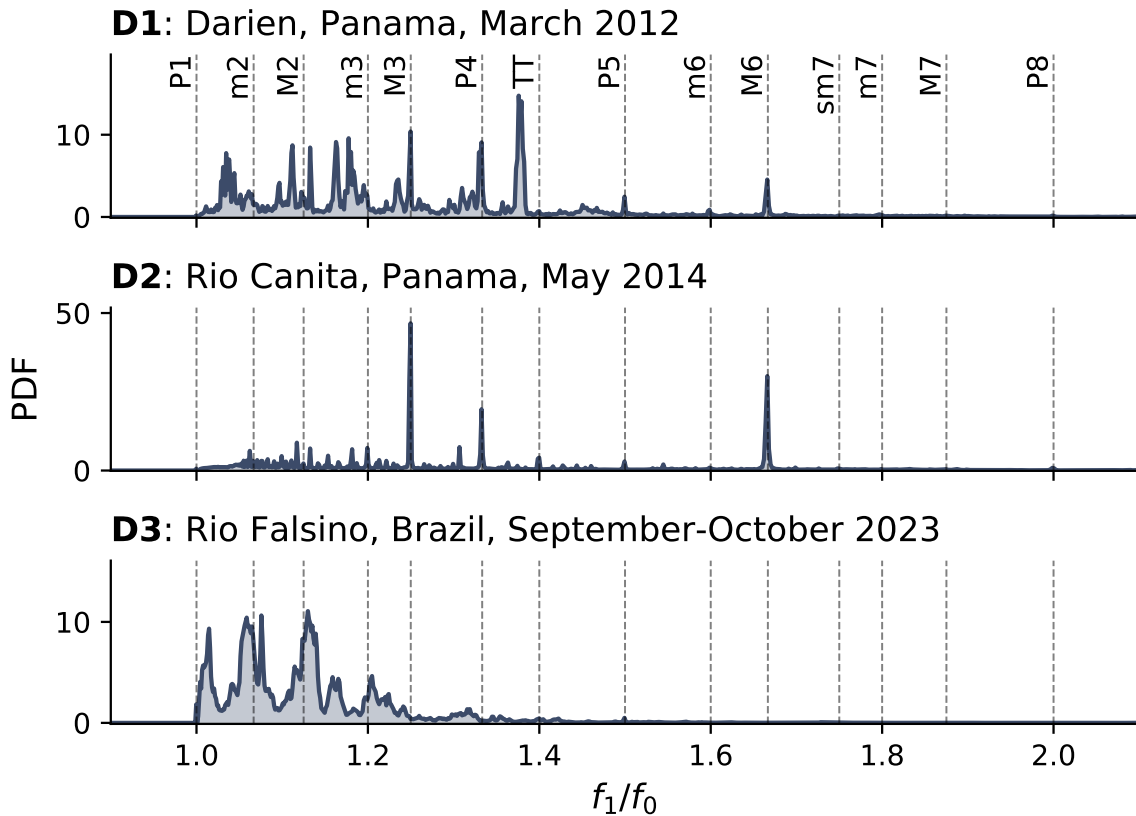


Figure 1: Distribution of electric organ discharge frequency ratios in different habitats. The electric organ discharge (EOD) frequencies of individual wave-type fish from different datasets recorded in their natural habitat using multi-electrode arrays were extracted. For every 10-second time window, the two strongest signals at each electrode were determined and the ratio of their EOD frequencies f_1/f_0 was calculated. Depicted is the distribution (PDF) of assumed frequency ratios. D1 = Dataset 1; D2 = Dataset 2; D3 = Dataset 3.

during the onset of rainy season in Darien, Panama, the most prominent peaks occurred at ratios matching the major third, perfect fourth, and major sixth (Figure 1, top). Smaller peaks corresponding to musical intervals appeared at the perfect fifth and minor sixth. The majority of non-musical frequency ratios assumed by fish fell between the unison and the minor third. Similarly, in Dataset 2 (D2) obtained during rainy season in the Río Cañita, Panama, large peaks were present at the major third, perfect fourth, and major sixth, accompanied by smaller peaks at the minor third, tritone, and perfect fifth (Figure 1, middle). As in D1, most remaining ratios were concentrated between the unison and the minor third. In both datasets, peaks at musical intervals were oftentimes larger in magnitude and more sharply defined than peaks appearing at non-musical ratios.

To quantify how precisely fish matched musical intervals, the mean deviation ($\bar{\Delta}r$), the standard deviation (σ_r), and the normalized standard deviation (σ_{norm}) for all fish pairs assuming a frequency ratio within a $\pm 1\%$ tolerance window relative to each musical interval were calculated (Figure 2, A and B, respectively). The normalized standard deviation allows comparisons between intervals, whereas the raw standard deviation can be compared to the mean deviation within each interval. Mean deviations smaller than the standard deviation indicate that fish clustered tightly around the exact musical interval, whereas mean deviations larger than the standard deviation suggest clustering near the border of the tolerance window.

Across most intervals, σ_{norm} was close to 1.0, that is, comparable to a normal distribution. Notable exceptions included the unison and octave, which exhibited smaller normalized standard deviations in all datasets ($\sigma_{norm} = 0.4444 - 0.5264$ and $0.5608 - 0.7897$ for the unison

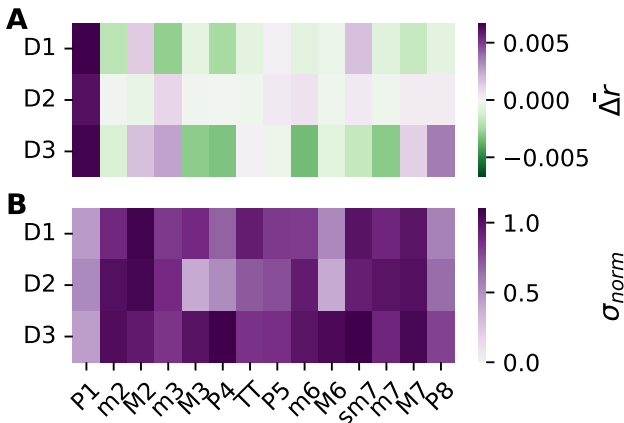


Figure 2: Deviation of assumed frequency ratios from musical intervals. Heatmaps depicting deviations of frequency ratios assumed by pairs of fish from target musical intervals within a $\pm 1\%$ tolerance window. Columns correspond to different musical intervals, and rows show data from D1 – D3 (Table 2, Figure 1). (A) Mean deviation $\bar{\Delta}r$ of fish pairs from musical intervals (B) Standard deviation σ_{norm} of fish pairs from musical intervals, normalized by the standard deviation of a uniform distribution of the same width to allow comparisons across intervals.

and octave, respectively). However, mean deviations for these intervals were substantially larger than their standard deviations, indicating that fish clustered at ratios far from the exact 1:1 and 2:1 ratios.

Other intervals with a small σ_{norm} appeared in D1 and D2. In both datasets, the normalized standard deviation was relatively small at the major sixth ($\sigma_{norm} = 0.5346$ and 0.3897 for D1 and D2, respectively), and mean deviations were smaller than standard deviations ($\sigma_r = 0.0051$, $\bar{\Delta}r = 0.0004$ and $\sigma_r = 0.0037$, $\bar{\Delta}r = 0.0008$ for D1 and D2, respectively). In D1, a similar but slightly looser clustering was observed at the perfect fourth ($\sigma_{norm} = 0.6906$, $\sigma_r = 0.0053$, $\bar{\Delta}r = 0.0025$). D2, on the other hand, showed very tight clustering around the major third ($\sigma_{norm} = 0.3912$, $\sigma_r = 0.0028$, $\bar{\Delta}r = 0.0001$) and perfect fourth ($\sigma_{norm} = 0.5220$, $\sigma_r = 0.0040$, $\bar{\Delta}r < 0.0001$).

In contrast, Dataset 3 (D3) collected during autumn 2023 in Brazil lacked distinct peaks at musical intervals as observed in the other two datasets (Figure 1, bottom). Instead, very broad peaks of frequency ratios appeared around the minor second, major second, and minor third. None of these peaks aligned precisely with any musical interval, as reflected by large normalized standard deviations ($\sigma_{norm} = 0.8364 - 1.1052$, excluding the unison and octave). Mean deviations were only small for intervals around which the distribution (Figure 1) was low, such as the tritone and perfect fifth. Furthermore, the overall width of the distributions differed markedly between D1 and D2. While fish in Panama assumed frequency ratios of up to the octave, the distribution computed from data collected in Brazil flattens beyond a ratio of 1.5. This indicates that overall difference frequencies were smaller as compared to Panama.

Together, these observations suggest that the formation of musical intervals in *Eigenmannia* may be influenced by environmental factors such as season or local habitats, with clear peaks at musical intervals appearing primarily during courtship season in Panama.

5.2 New Low Frequencies Arise Near Musical Intervals

To explore whether the preference for musical intervals could arise from inherent spectral differences between musical and non-musical frequency ratios, simulated superpositions of *Eigenmannia* waveforms were analyzed. Due to the prominence of the minor third, perfect fourth, and perfect fifth in the field data (Figure 1), the analysis focused on these intervals.

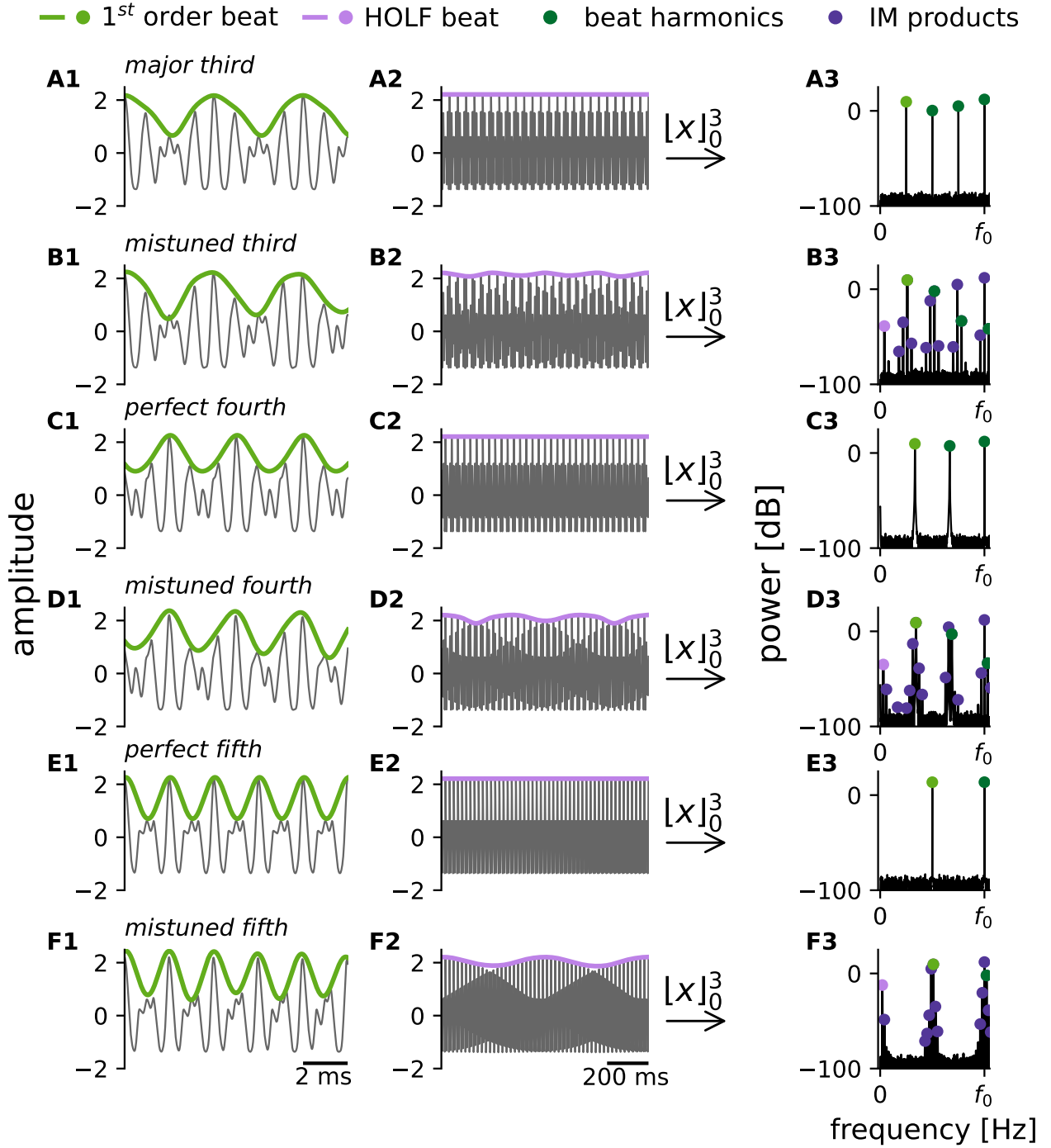


Figure 3: Spectral differences between musical and mistuned intervals. Depicted are simulated superpositions of two *Eigenmannia* waveforms constituting a major third (A), perfect fourth (C), perfect fifth (E), and corresponding mistuned intervals (B, D, and F, respectively, $\$epsilon = 0.01$). For each interval, column 1 shows the raw waveform on a short timescale together with the resulting first order beat at the difference frequency (light green). Column 2 displays the same signals on a longer timescale, in which higher-order low-frequency (HOLF) beats become apparent (light purple). Column 3 shows the power spectra of thresholded of cubed versions of the signals, with spectral peaks being labeled according to whether they correspond to the beat frequency and its harmonics (light and dark green) or to HOLF components and other intermodulation products (light and dark purple).

For each examined musical interval (Figure 3, A, C, and E), a slightly mistuned version ($\epsilon = 0.01$, Figure 3, B, D, and F) was selected to compare potential differences in the waveforms. First-order beats at the difference frequency Δf are visible on all intervals when viewed on small time windows (20 ms, Figure 3, column 1). The structure and frequency of first-order beats on mistuned intervals closely resembles their musical counterparts. In a zoomed-out view (2 s, Figure 3, column 2), however, clear differences emerge. Mistuned intervals exhibit additional amplitude modulations on top of the first-order beat, in the following referred to as higher-order low-frequency (HOLF) beats. These modulations arise from non-linear interactions between harmonics of the underlying signals (Figure 3, column 2, purple line). These HOLF beats remain absent at musical intervals, as many of their harmonics align exactly.

In order to generate additional peaks corresponding to the frequency of amplitude modulations in the spectral domain, the combined waveforms must pass through a non-linearity. We applied a cubed threshold function, which resembles the non-linearity of P-unit electroreceptors (Barayeu et al. 2023). Power spectra of the thresholded signals reflect the differences between musical and mistuned intervals observed in the temporal domain (Figure 3, column 3). While musical intervals only produce peaks at the beat frequencies and their harmonics, mistuned intervals give rise to additional peaks, known as intermodulation (IM) products (purple dots). These IM components cluster both in the low-frequency range and around the peaks that are also present in the spectra of musical intervals.

To quantify spectral differences across frequency ratios within the octave, the number of spectral peaks at each frequency ratio was counted (Figure 4, A). For most ratios, the peak count remains relatively stable (70 – 73 peaks). At most musical intervals, however, the number of spectral peaks sharply decreases, indicating that these intervals generate comparatively simpler spectra. The only exceptions to this trend are the minor second, major second, and major seventh, all of which do not change in peak count more than baseline and are among the most dissonant intervals as reported in perceptual studies (Bowling et al. 2018; McDermott et al. 2010). Another notable deviation from this pattern occurs at $\frac{f_1}{f_0} = 1.375$, where 75 peaks have been counted in the spectrum. This interval, which is located between the perfect fourth and tritone and sometimes referred to as an undecimal augmented fourth, is the only one whose peak count increases above baseline.

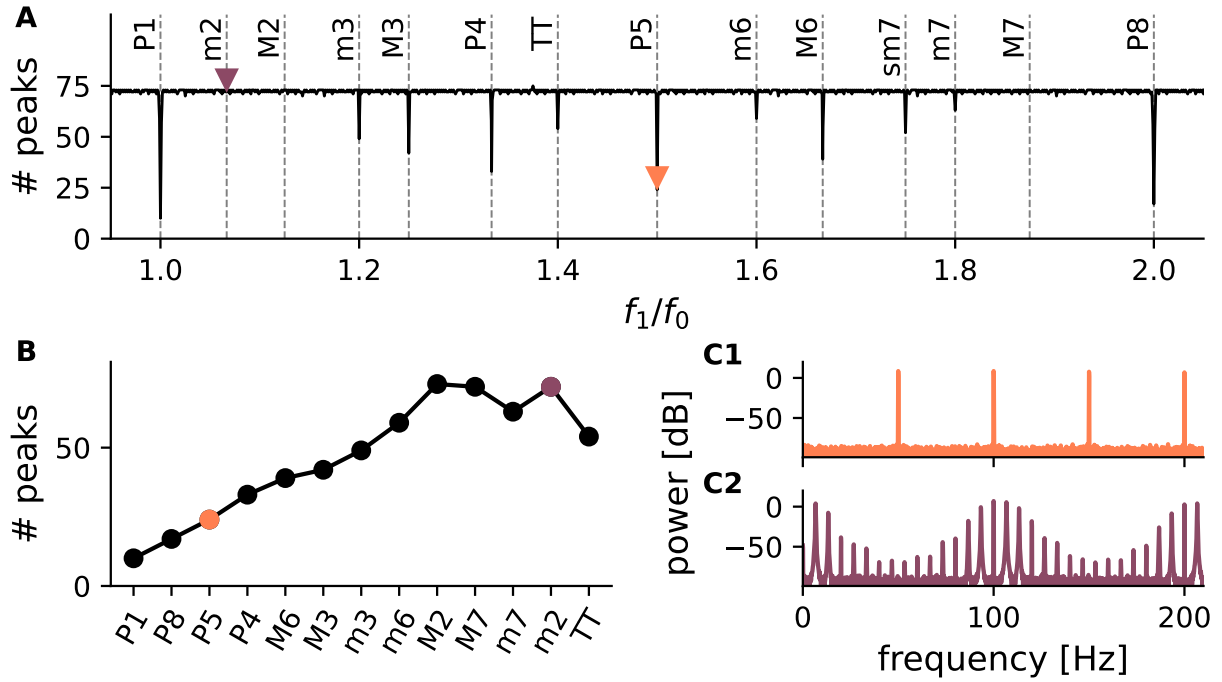


Figure 4: Number of peaks across different intervals. (A) Number of spectral peaks detected in the amplitude spectra for all intervals within one octave (step size = 0.001). Ratios corresponding to musical intervals are indicated by dotted lines. (B) Number of peaks as function of perceived consonance of musical intervals. Consonance rankings were obtained from Bowling et al. (2018), McDermott et al. (2010), Shapira Lots and Stone (2008), and Trulla et al. (2018). The subminor seventh was excluded from this depiction due to a lack of reported ratings for this interval. (C) Exemplary power spectra of a perfect fifth (C1), a highly consonant interval, and a minor second (C2), one of the most dissonant intervals. Their positions in A and B are highlighted in purple and orange, respectively. Abbreviations for musical intervals are provided in Table 1.

Because previous research suggests a possible link between properties that increase spectral complexity and perceived pleasantness of a musical interval (Helmholtz 2009; Smit et al. 2019), the relationship between peak count and perceptual ratings was examined using consonance judgements reported by Bowling et al. (2018), McDermott et al. (2010), Shapira Lots and Stone (2008), and Trulla et al. (2018) (Figure 4, B). Up to the major second, the number of detected peaks increases as perceived consonance decreases. For the most dissonant intervals (major seventh, minor seventh, minor second, and tritone), this relationship flattens, with most of these intervals exhibiting similarly high peak counts. Small decreases at the tritone and minor seventh are visible, which might have been influenced by these ratios being differently defined in different tuning systems.

Directly comparing the spectra generated by a highly consonant (perfect fifth, Figure 4, C1) and highly dissonant (minor second, Figure 4, C2) interval further highlights these differences. The minor second produces a large amount of IM products, thereby exhibiting many additional spectral peaks, while the perfect fifth only generates peaks at frequencies associated with the first-order beat and its harmonics. Similarly to the mistuned intervals examined in Figure 3, dissonant intervals are characterized by the emergence of additional frequency components. However, in contrast to mistuned intervals, these components appear more orderly and regularly spaced in the minor second.

To further investigate the occurrence of HOLF beats across frequency ratios, the standard deviation σ_{sig} of signals convolved with either a narrow or broad kernel was computed. This measure reflects the magnitude of high- and low-frequency modulations, respectively. For the signals convolved with the narrow kernel, σ_{sig} is largest around the unison, and declines steadily until $f_1/f_0 = 1.5$. For ratios above the perfect fifth, σ_{sig} increases again slightly, before dropping off after the octave. This pattern indicates that higher-frequency AMs are strongest around the unison and octave, and weakest around the perfect fifth. Notably, σ_{sig} increases sharply at the perfect fourth, perfect fifth, and major sixth. For the broad kernel, which only preserves HOLF beats, σ_{sig} is high for ratios around the unison and octave, and shows a smaller rise around the perfect fifth. At exactly the corresponding musical intervals, however, σ_{sig} drops off sharply. This indicates that HOLF beats are strongest for mistuned unisons, mistuned octaves, and mistuned fifths, but disappear at exact musical intervals. This finding is congruent with the

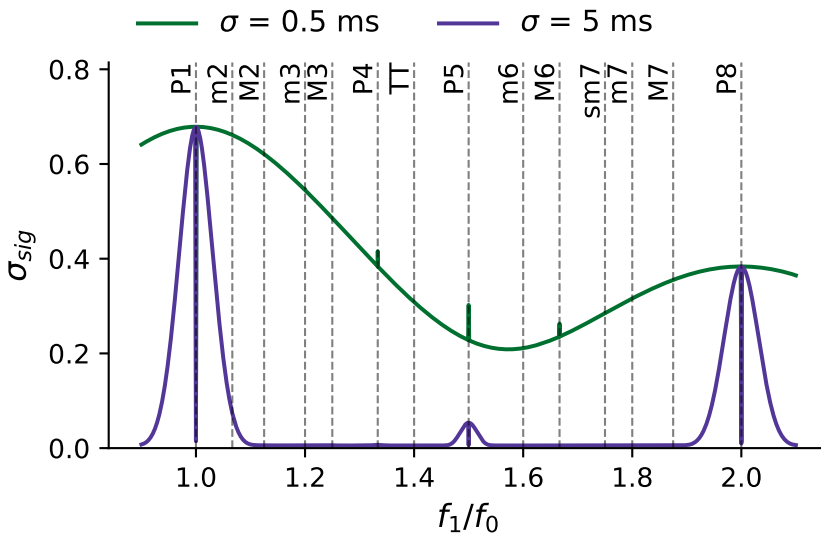


Figure 5: Standard deviation of convolved signals. The standard deviation of convolved signals computed using either a narrow ($\sigma = 0.5$ ms, green) or broad ($\sigma = 5$ ms, purple) kernel. Ratios corresponding to musical intervals are indicated by dotted lines. Abbreviations are explained in Table 1.

observations from Figure 3. At all other frequency ratios, however, σ_{sig} remains low, indicating that HOLF beats, if present, are only of small magnitude.

Conclusively, these analyses reveal systematic spectral differences between musical and non-musical frequency ratios. Musical intervals are characterized by comparatively simple spectra with few IM components. Non-musical intervals, on the other hand, are characterized by a large number of IM products and therefore more complex spectra. For ratios near musical intervals, that is, mistuned intervals, prominent IM products occur in low frequency regimes, giving rise to HOLF beats. Importantly, these HOLF beats and other higher-order frequency components can be generated through a single non-linearity at the level of individual cells, and does not depend on computations based on frequency decomposition as found in the cochlea.

5.3 HOLF Beats are Represented in the Electrosensory System

Given weakly electric fish align themselves to musical intervals that are characterized by the *absence* of HOLF beats, the question arises whether such slow higher-order modulations are represented within the electrosensory system. To address this, the responses of 39 P-unit model cells fitted to electrophysiologically recorded P-units (<https://github.com/bendalab/punitmodel>, Barayeu et al. 2023) to stimuli with frequency ratios close to, or exactly at, musical intervals were characterized.

The spike trains generated by model cells differ systematically in response to musical and mistuned intervals (Figure 6). At both types of intervals, the spiking pattern reflects the fast-first order beat. However, mistuned intervals produce spike trains that are additionally modulated by the HOLF beat, with spiking activity ceasing at fixed phases of the slow modulation (Figure 6, A and B). To examine how these responses would be represented at different timescales of temporal integration, spike trains were convolved with Gaussian kernels of two different widths ($\sigma = 0.5$ ms and $\sigma = 5$ ms; Figure 6, C, top and bottom, respectively). With the narrow kernel, the convolved firing rate closely follows the structure of the first-order beat in response to both the musical and mistuned ratio, with an additional slow modulation mirroring the HOLF beat at the mistuned interval. The broad kernel, in contrast, effectively filters out the fast-first order beat, producing a steady firing rate in response to the musical interval. However, the slow modulation corresponding to the HOLF beat remains visible in response to mistuned

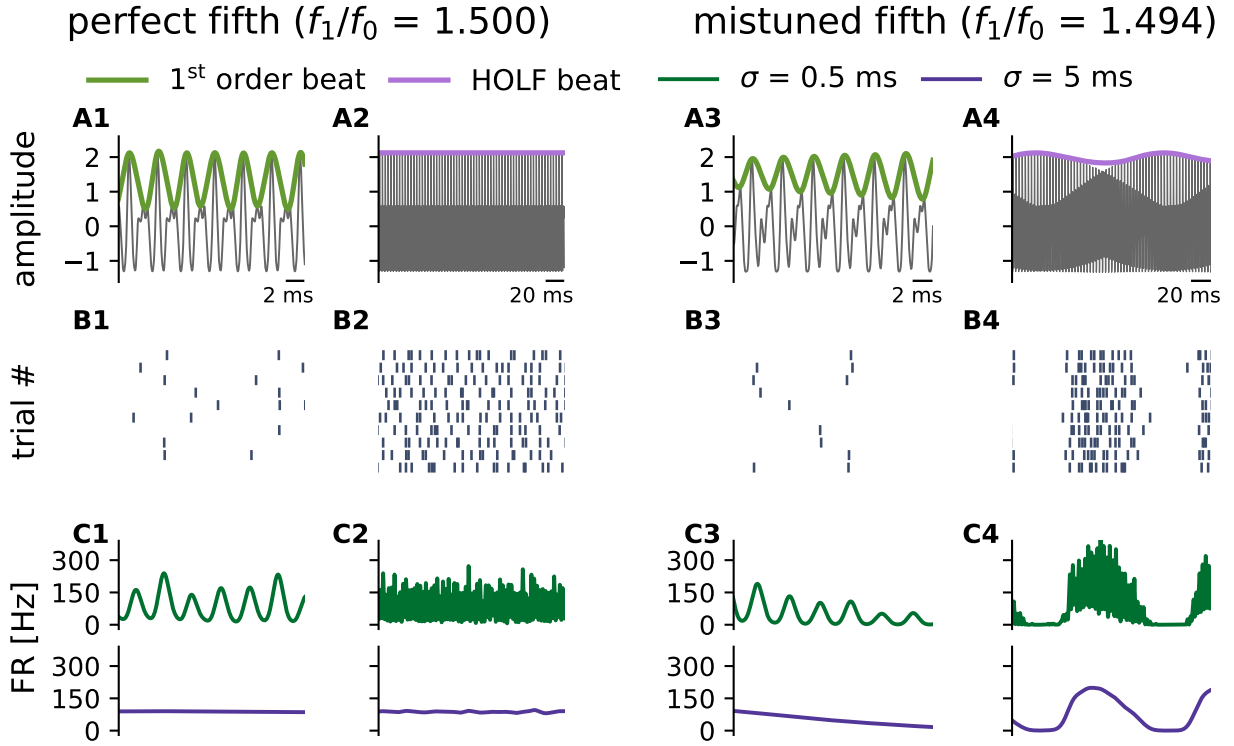


Figure 6: Responses of a model cell to a perfect and mistuned fifth. (A) Exemplary stimuli used to drive model cell responses. Shown are a perfect fifth ($f_1/f_0 = 1.500$, **A1**, **A2**) and mistuned fifth ($f_1/f_0 = 1.494$, **A3**, **A4**). Each stimulus is presented on both a zoomed-in (20 ms, **A1**, **A3**) and a more extended view (2s, **A2**, **A4**) to illustrate different orders of beats. First-order beats are traced in light green, higher-order low-frequency (HOLF) beats in light purple. (B) Spike raster plots depicting responses of the same model cell to stimuli shown in A. (C) Convolved firing computed from spike trains in B. The upper traces (green) were generated using a narrow kernel ($\sigma = 0.5$ ms), whereas the bottom traces (purple) were convolved with a broader kernel ($\sigma = 5$ ms).

intervals even at this longer integration timescale. This indicates that HOLF beats are robustly preserved across integration windows, whereas fast first-order beats vanish when information is integrated over longer timescales.

While the convolved firing rates reveal how different orders of amplitude modulations appear at different integration timescales, power spectra can reveal frequency components dominating the neural response. For all examined musical intervals (Figure 7, A1 – C1; major third - purple; perfect fourth - pink; perfect fifth - orange), the cell response is dominated by the fast first-order beat and its harmonics. Slightly mistuning each musical interval by a factor $\epsilon = 0.005$ (Figure 7, A2 – C2) introduces additional frequency components in the power spectrum, clustering both in the low-frequency regime as well as around the first-order beat peaks. This structure parallels the pattern observed in spectra of the thresholded signals (Figure 3). Such similarities were

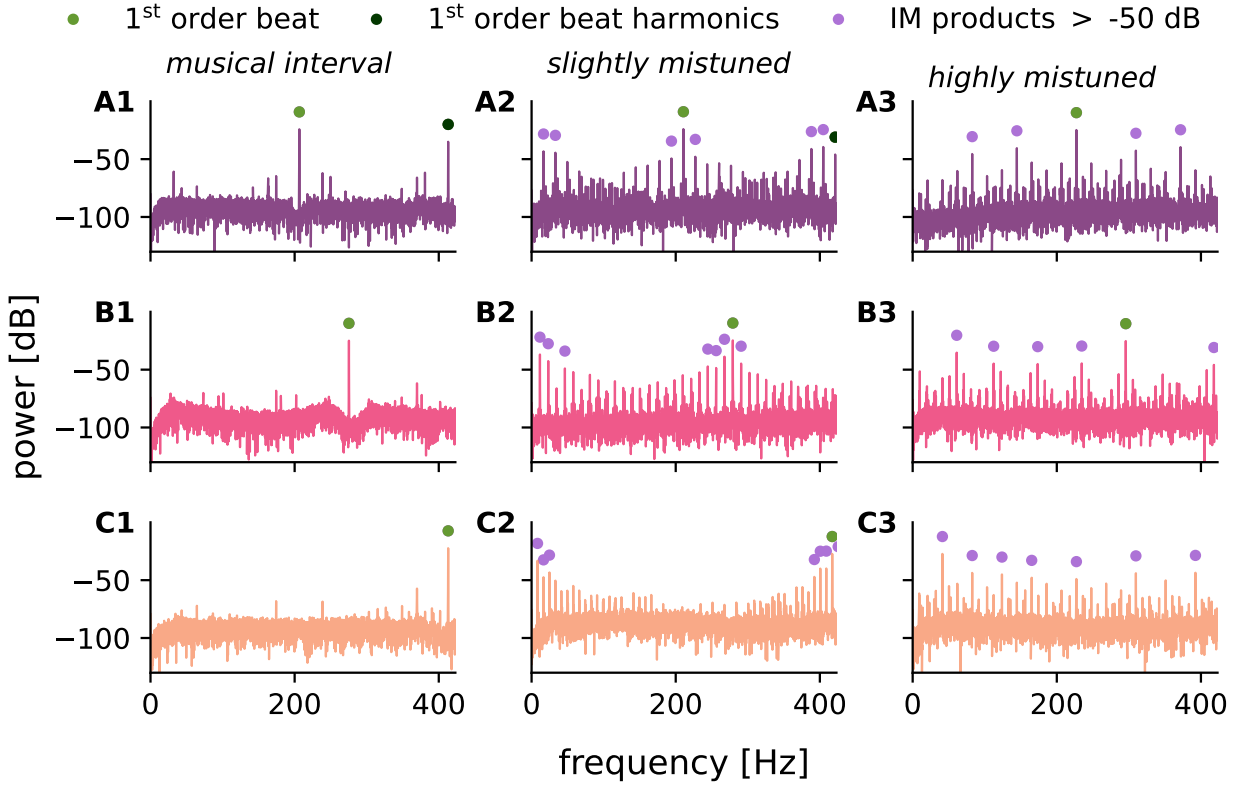


Figure 7: Powerspectra of cell responses to musical and mistuned intervals. Powerspectra computed from spikes trains of the model cell depicted in Figure 6. **A1 – C1** show responses to musical intervals. Mistuned intervals were generated by adding a mistuning factor ϵ to the ratio r_{mi} of the corresponding musical interval. For slightly mistuned intervals (**A2 – C2**), $\epsilon = 0.005$, and for highly mistuned intervals (**A3 – C3**), $\epsilon = 0.025$. Rows correspond to different intervals: **(A)** Major and mistuned thirds (purple). **(B)** Perfect and mistuned fourths (pink). **(C)** Perfect and mistuned fifths (orange). Peaks at the first-order beat frequency are highlighted in green, and those corresponding to intermodulation (IM) products with a power greater than -50 dB in purple.

expected, as the model contains the same cubed threshold non-linearity that has been applied to the input signal. Importantly, this pattern persists despite the additional non-linearities imposed by the spike generation mechanism, indicating that the LIF model does neither distort nor abolish higher-order frequency components. With stronger mistuning ($\epsilon = 0.025$), the clusters of IM product peaks break apart, producing a more regular pattern of peaks shifted toward higher frequencies. No prominent components below 40 Hz remain, indicating that larger mistunings no longer generate low-frequency components in the neural response.

To summarize modulation strengths for all cells and ratios within the octave, we computed the standard deviation σ_{FR} of the convolved firing rates. σ_{FR} is expected to be low for steady response patterns, and increases with the strength of firing rate modulation. Averaged over

all 39 cells, the modulation depths of the firing rates as a function of stimulus frequency ratio follows markedly different patterns for the two kernel widths (Figure 8). For the narrow kernel ($\sigma = 0.5$ ms, high-opacity green), the mean σ_{FR} decreases steadily between the minor second and major sixth, and then rises again towards the octave. At most musical intervals, small dips of σ_{FR} can be observed, indicating a decreased response to high-frequency amplitude modulations in the signal. Near the unison and the octave, σ_{FR} increases before dropping sharply at the interval itself. This indicates that the firing rate modulation at small timescales of temporal integration was strongest for mistuned unisons and octaves, and decreases for all ratios in between, most strongly at musical intervals. This pattern largely mirrors that of the standard deviation σ_{sig} of the convolved signal (low-opacity green). However, the broad minimum around the perfect fifth is slightly elevated for σ_{FR} compared to σ_{sig} . Furthermore, in contrast σ_{FR} , σ_{sig} exhibits sharp upward peaks at the perfect fourth, perfect fifth, and major sixth.

For the broad convolution kernel ($\sigma = 5$ ms, high-opacity purple), σ_{FR} is generally lower than for the narrow kernel. However, around each musical interval except the major seventh, σ_{FR} increases sharply, followed by a minimum at the exact musical interval. This pattern closely resembles the pattern around the unison and octave for the narrower kernel. This indicates that slow modulations of the firing rate are strongest for slightly mistuned intervals, but weakest at exact musical intervals. Moreover, the magnitude of the modulation at mistuned intervals appears to be ordered according to the pleasantness of the nearby musical interval. It is

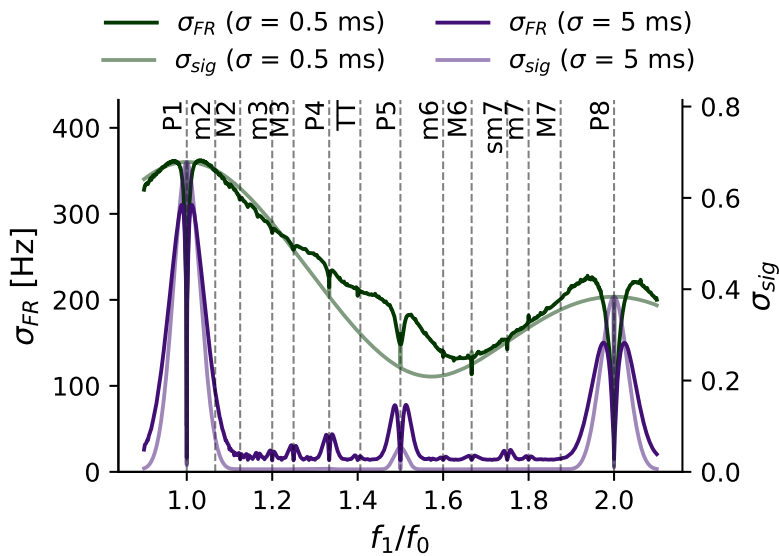


Figure 8: Mean deviation of convolved firing rates. The mean standard deviation of the convolved firing rate over all 39 model cells computed using two different kernel widths ($\sigma = 0.5$ ms, green and $\sigma = 5$ ms, purple). Low-opacity lines depict the standard deviation of the convolved thresholded signals as already shown in Figure 5. Ratios corresponding to musical intervals are indicated by dotted lines.

strongest for the unison and octave, followed by the perfect fifth, perfect fourth, and major third. Comparing this pattern to the standard deviation of the input signal convolved with the same kernel width σ_{sig} (low-opacity purple) reveals notable differences. Increases in σ_{sig} are only observed around the unison, octave, and perfect fifth, while σ_{FR} increases near most musical intervals. Furthermore, while σ_{FR} is lower at the unison and octave than σ_{sig} , its magnitude increases relative to σ_{FR} at the perfect fifth. This indicates that non-linearities introduced by the spike generation mechanism of the LIF model not only preserve, but even amplify HOLF beats relative to the thresholded signal.

Overall, these findings show that the structure of HOLF beats is preserved across different integration windows, whereas fast first-order beats persist only at short integration timescales. Importantly, the non-linearities of the spike generator mechanism within the LIF model amplify HOLF beats compared to the thresholded input signal, suggesting that the electrosensory system enhances slow higher-order modulations produced by mistuned ratios.

5.4 Stimuli Near Musical Intervals Elicit Behavioral Responses

So far, the presented results demonstrate that mistuned intervals generate HOLF beats that remain absent in musical intervals. Moreover, our findings strongly suggest that these HOLF beats are not only represented in the electrosensory system, but even amplified by additional non-linearities in the spike generation mechanism of the LIF model. Hence, it is plausible to assume that fish are able to perceive HOLF beats produced by mistuned intervals. We tested this prediction using the well-studied Jamming Avoidance Response (JAR). In the JAR, fish shift their EOD frequency when encountering stimuli that generate low-frequency first-order beats, likely to avoid interference with electrolocation (Heiligenberg et al. 1978; Watanabe and Takeda 1963). Thus, we reasoned that fish should likewise demonstrate a JAR-like frequency adjustment in response to HOLF beats. This prediction was tested in a classic JAR paradigm in which different musical and mistuned intervals were presented to the fish.

Mistuned unisons, corresponding to classical JAR stimuli, frequently evoked clear JARs in all fish, characterized by a quasi-exponential frequency shift at stimulus onset followed by a return to baseline once the stimulus ceased (Figure 9, A1). Indeed, mistuned versions of other musical intervals were able to induce JAR-like frequency shifts in most subjects as well. However,

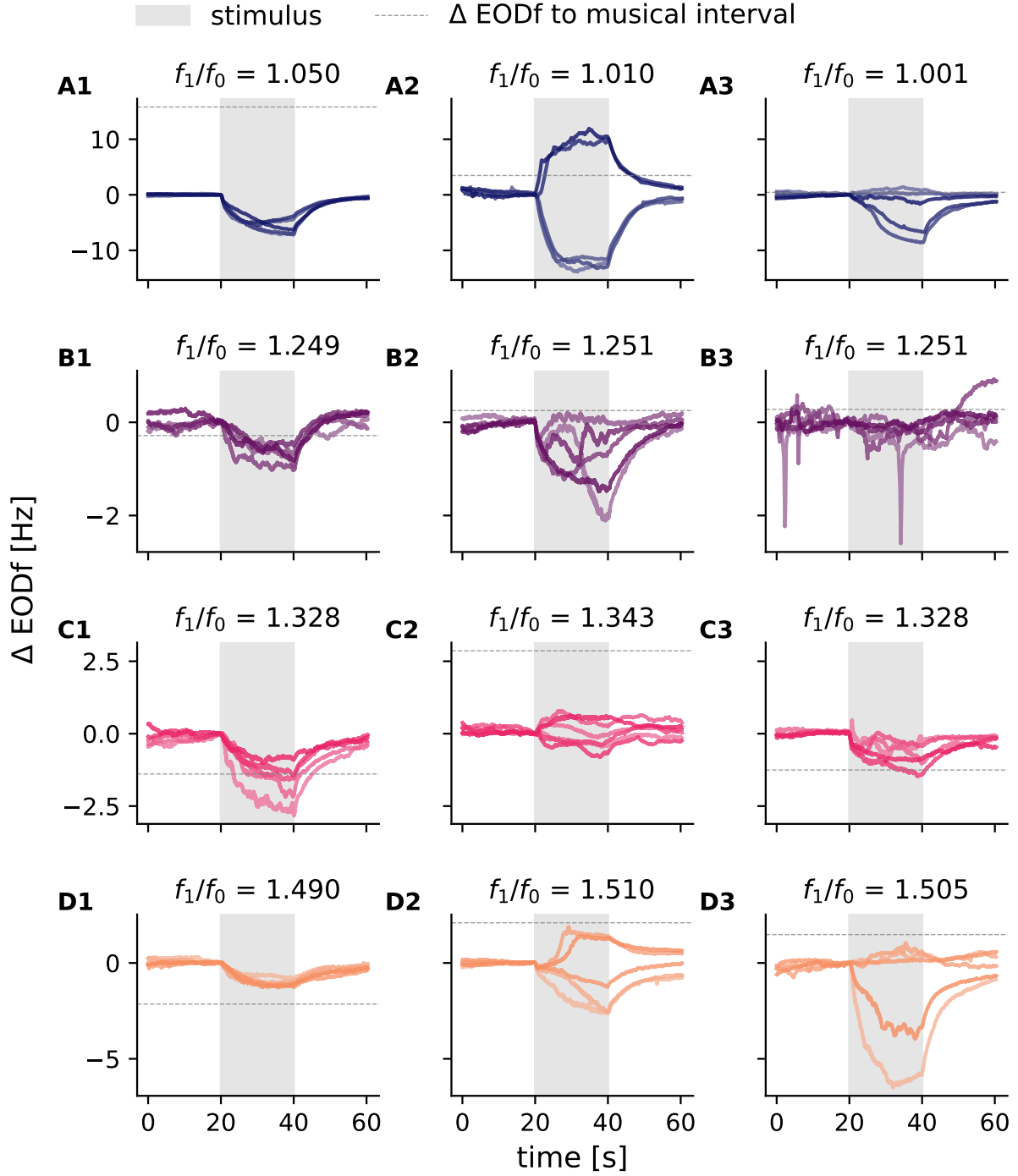


Figure 9: Exemplary responses to multiple repeats of the same stimulus. Time courses of EOD frequency changes for multiple repeats of the same stimuli. Changes are shown relative to each fish's baseline EOD frequency before stimulus presentation (Δ EODf). Stimuli comprise mistuned versions of the unison (A, blue), major third (B, purple), perfect fourth (C, pink), and perfect fifth (D, orange), with the exact frequency ratio between stimulus (f_1) and fish (f_0) indicated above each panel. Differently colored lines represent responses to different repeats, and each panel depicts the responses of the same fish upon repeated presentations of the same stimulus. The gray background indicates the time window during which the stimulus was presented.

responses varied strongly across stimuli and individual fish. Figure 9 illustrates this variability through exemplary responses to repeated presentations of identical stimuli.

Whereas the unison condition typically induced frequency shifts away from a 1:1 ratio, the direction of frequency shifts in response to other mistuned intervals was less consistent. Some fish reliably moved toward the musical interval (Figure 9, A1 – D1), but several fish also showed substantial trial-to-trial variability in the the direction of their frequency shifts. Especially ratios very close to a musical interval, including near the unison, triggered both upwards and downwards shifts across different trials (Figure 9, A2, C2, D3). In some cases, fish even displayed both up- and downwards shifts within a single stimulus presentation (Figure 9, B2) before stabilizing the direction of their EOD frequency shift. Some fish only responded to a subset of the presented repeats, (Figure 9, A3 and D3), whereas others did not seem susceptible to certain stimuli at all (Figure 9, B3). Some fish oftentimes displayed a small and rapid upward deflection at stimulus onset, potentially producing a chirp before the main frequency shift (Figure 9, C3). Overall, these observations demonstrate that responses varied substantially not only between fish, but also within individual fish across identical stimulus repetitions.

To summarize the response behavior of each fish, median responses were computed for all tested ratios. Responses to stimuli around the unison displayed the largest variability across fish in both the low- (Figure 10, A, blue) and the high-contrast (Figure 10, B, blue) condition. Frequency shifts ranged from approximately -20 Hz to +20 Hz, although the majority of fish clustered in a range from -7 Hz to +6 Hz. Some individuals shifted their EOD frequency almost exclusively upwards, independent of stimulus ratio. Two fish exhibited markedly larger frequency shifts in both the low- and high-contrast condition compared to the other fish, whereas one fish consistently decreased its EOD frequency by about -7 Hz for all ratios tested in the high-contrast condition. Based off classical JAR literature (Heiligenberg et al. 1978; Watanabe and Takeda 1963), fish were expected to increase their EOD frequency for all ratios below 1.0, and decrease it for ratios above 1.0. However, this pattern did not appear. Instead, many fish displayed a consistent bias toward either upward or downward shifts across most ratios. Only stimuli with a ratio of 1.05 led to consistent downward shifts across all fish. Response magnitudes showed no relationship to the size of σ_{FR} at the corresponding frequency ratios (Figure 10, C, blue).

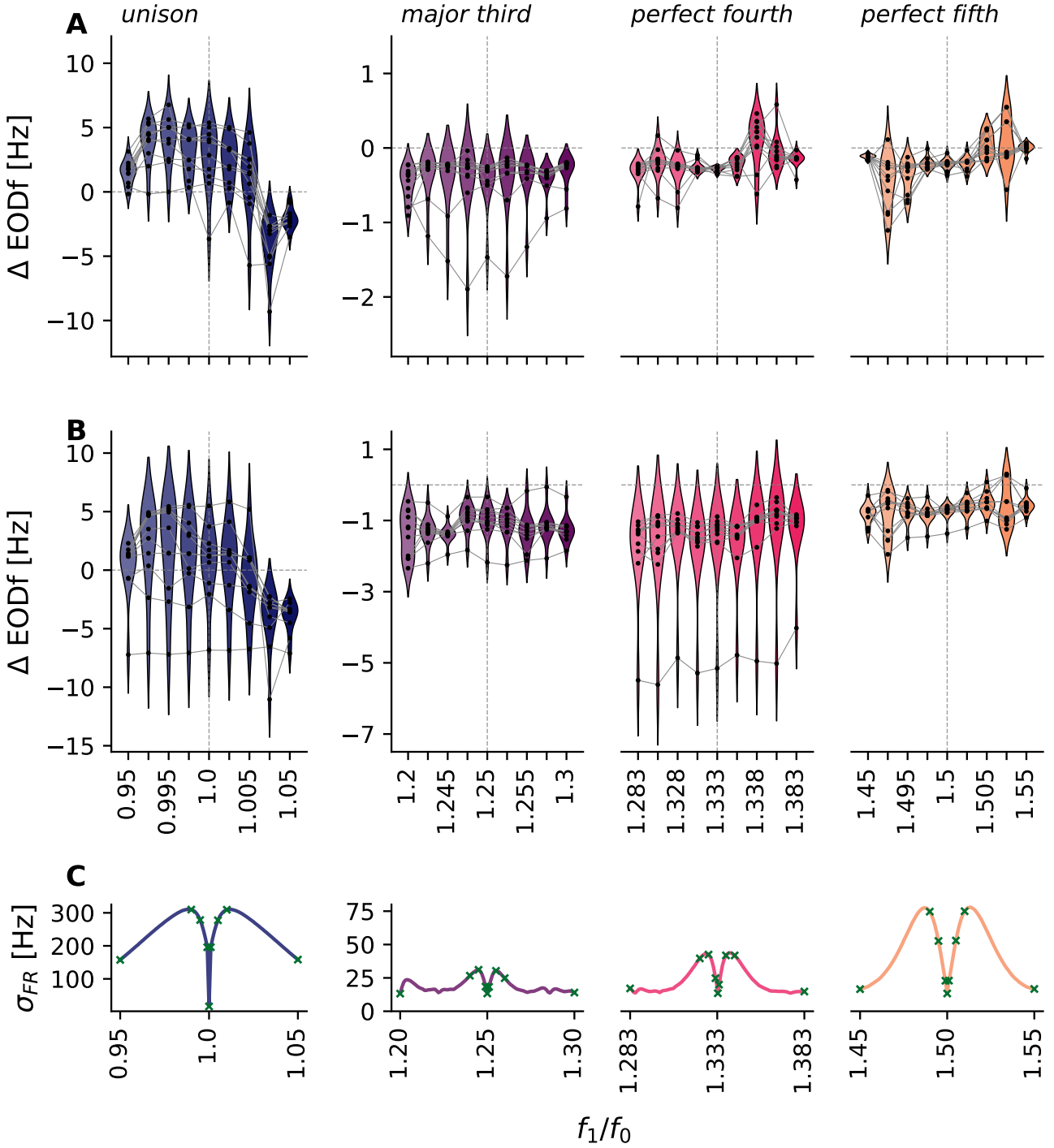


Figure 10: Median response of all fish for each tested frequency ratio. [LS: add what colors represent] Relative change in EOD frequency (ΔEODf) from baseline during stimulus presentation for the low contrast (**A**, 80%) and high contrast (**B**, 300%) condition. Dots represent the median response of each fish, averaged across all repeats of the same stimulus. Points corresponding to the same individual are connected by thin grey lines. Zero change is indicated by the horizontal dotted line, while the stimulus corresponding to the exact musical interval is highlighted by the vertical dotted line. (**C**) Standard deviation of convolved firing rate σ_{FR} for the range of frequency ratios tested in the JAR experiments. Due to the different scaling of the x-axes, ratios corresponding to the stimuli tested in **A** and **B** have been highlighted by green crosses.

Compared to the unison condition, response magnitudes for all musical intervals were markedly diminished, ranging from approximately -2 Hz to +2 Hz in the low-contrast (Figure 10, A), and -4 Hz to +4 Hz in the high-contrast (Figure 10, B) condition. In response to different mistuned versions of the major third (Figure 10, A and B, purple), most fish decreased their EOD frequency, although the magnitude of the shifts was considerably variable. One fish consistently produced larger downwards shifts in both the low- and high-contrast condition than the other fish. Again, the shape of σ_{FR} (Figure 10, C, purple) did not correspond to the magnitude of responses across intervals. However, for this stimulus subset it needs to be noted that the range of tested frequency ratios did not only cover mistuned major thirds. Instead, it included ratios near the minor third at $f_1/f_0 = 1.2$ and the septimal major third at $f_1/f_0 = 1.285$. Around these ratios, σ_{FR} exhibits the typical increase, indicating the presence of HOLF beats, before dropping sharply at the exact intervals.

Responses to low-contrast stimuli near the perfect fourth were even more smaller than for the major third (Figure 10, A, pink). Frequency shifts seldomly exceeded ± 1 Hz. High-contrast stimuli, on the other hand, elicited responses of similar magnitude than those around the major third (Figure 10, B, pink). The variability of responses in the low-contrast condition strongly depended on the stimulus ratio, being largest for $\epsilon = \pm 0.005$ and $\epsilon = \pm 0.01$. These ratios correspond to the ratios generating the largest σ_{FR} within the tested range. Furthermore, at these ratios, the population did not exhibit a consistent direction of frequency shifts, whereas most fish decreased their EOD frequency for the remaining ratios. In the high-contrast condition, variability remained similar across tested ratios, and most fish reliably decreased their EOD frequencies irrespective of frequency ratio. Again, one individual exhibited notably stronger responses compared to the other fish.

Mistuned perfect fifths produced the strongest responses among musical intervals, excluding the unison. Response magnitudes ranged from -2 Hz to +2 Hz in the low-contrast (Figure 10, A, orange) and -4 to +4 Hz in the high-contrast (Figure 10 B, orange) condition. There was no strong tendency towards up- or downwards shifts in EOD frequency for the low-contrast condition, whereas most fish decreased their frequency in response to high-contrast stimuli. Again, the variability of responses in the low-contrast condition mirrored the shape of σ_{FR} in the same ratio range, peaking at $\epsilon = \pm 0.005$ and $\epsilon = \pm 0.01$. This relationship was markedly weaker

in the high-contrast condition. Again, certain individuals appeared to exhibit disproportionately larger responses than the remainder of tested subjects.

Because this work highlights the relevance of frequency ratios rather than absolute frequency differences, changes in frequency ratio were quantified as well. For mistuned unisons, fish generally shifted their EOD frequency away from the 1.0 ratio in both the low- (Figure 11, C, blue) and high-contrast (Figure 11, D, blue) condition. The previously noted tendency for many individuals to decrease their EOD frequency also for ratios slightly above 1.0 is also reflected by these data. Ratio changes near the major third were, on average, small for the low-contrast condition (Figure 11, C, purple), whereas an increase in ratio could be observed for all stimuli in the high-contrast condition (Figure 11, D, purple). In certain individuals, an interesting pattern for low-contrast stimuli near the perfect fourth and perfect fifth (Figure 11, A and B, pink and orange, respectively) occurred. Instead of shifting their EOD frequency in a way that increased the distance to the exact musical interval, these individuals displayed a consistent convergence toward the musical interval. However, this tendency did not appear at the population level because only a subset of individuals showed this behavior (Figure 11, C, pink and orange, respectively). In contrast, most fish decreased their EOD frequency when presented with the same frequency ratios at a high contrast, producing an increase in ratio (Figure 11, D, pink and orange, respectively).

Overall, these results demonstrate that fish produce JAR-like frequency shifts when exposed to mistuned musical intervals. However, the magnitude and direction of frequency shifts was highly variable across fish, and strongly depended on stimulus ratio and stimulus contrast. Mistuned unisons reliably produced strong JAR responses. For other musical ratios, the magnitude of the EOD frequency shift appeared to be related the magnitude of σ_{FR} at the corresponding ratio. Furthermore, a small subset of individuals consistently shifted their EOD frequency toward the exact musical interval, although this effect did not appear at the population level.

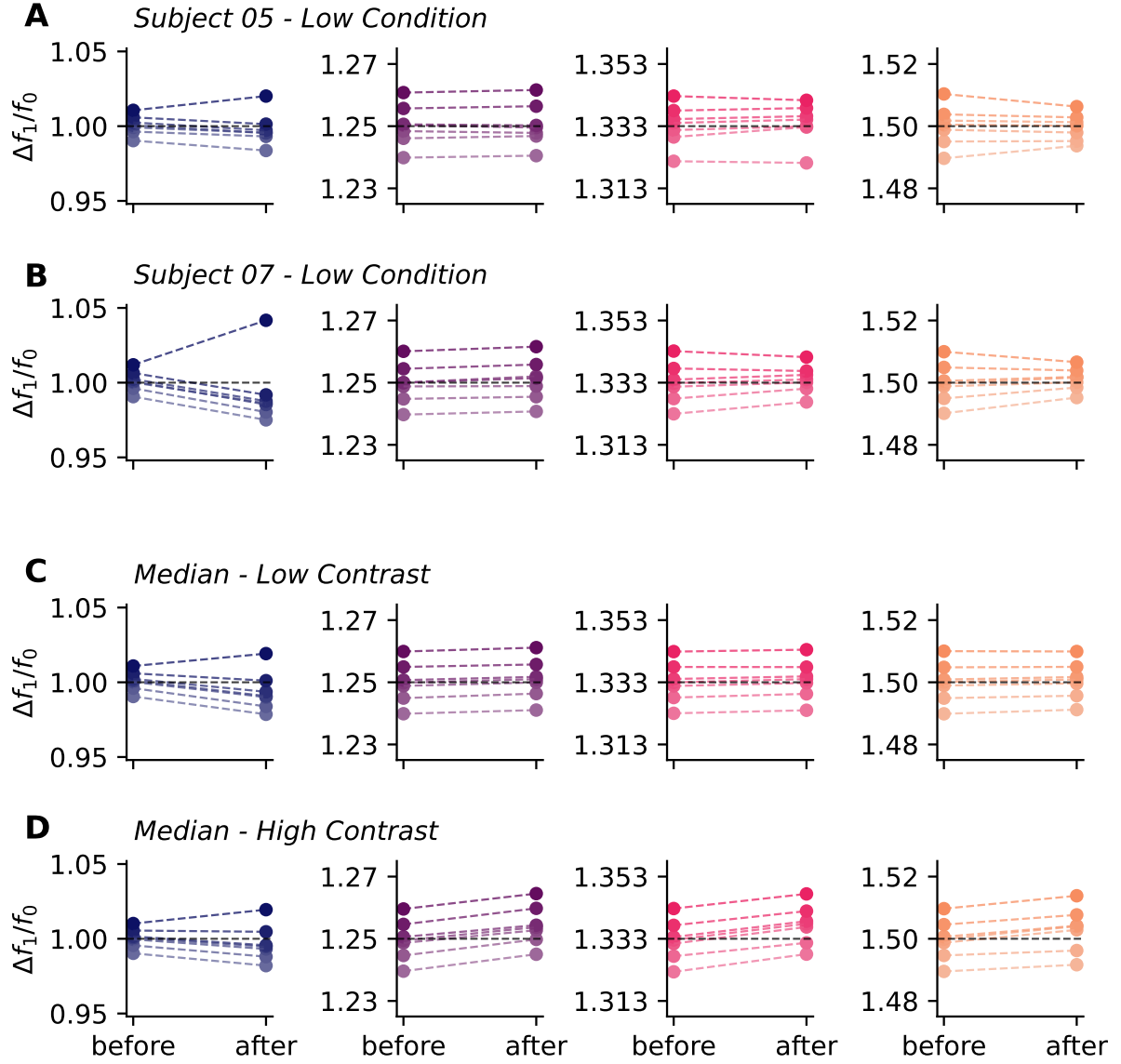


Figure 11: Median change in frequency ratio for different mistuned intervals. Frequency ratio between stimulus and fish at stimulus onset (before) and stimulus offset (after) in response to stimuli centered around the unison (blue), major third (purple), perfect fourth (pink), and perfect fifth (orange). (A), (B) Median change in ratio across low-contrast conditions (80%) for two individual fish. (C), (D) Median change across all individuals for the low-contrast and high-contrast conditions (300%), respectively.

6 Discussion

The reasons behind the differential percept of consonant and dissonant musical intervals have been debated for centuries (Di Stefano et al. 2022; Helmholtz 2009). Although many efforts have been made to identify the mechanisms underlying this phenomenon, no single theory accounts for it in its entirety yet. Psychophysical evidence is often ambiguous (Bidelman and Krishnan 2009; Bones et al. 2014; Cousineau et al. 2012), and studies including non-Western listeners have failed to demonstrate a reliable preference for consonant intervals across cultures (Maher 1976; McDermott et al. 2016). These findings have put forward the idea that the concepts of consonance and dissonance are largely culturally shaped.

The current work demonstrates that weakly electric fish display a bias towards musical intervals in a modality different from the auditory system. This unique model organism therefore allows us to approach the problem of consonance and dissonance from a new perspective. We found that basic non-linearities within individual receptor cells are sufficient to generate the spectral differences between musical and non-musical frequency ratios. Musical intervals are characterized by the absence of higher-order amplitude modulations, whereas 'mistuned' intervals generate additional low-frequency beats that result from the non-linear interactions between harmonics. Similarly, dissonant intervals are also distinguished from consonant intervals by increased power in the low-frequency domain originating from the non-linear interactions between non-overlapping harmonics. Furthermore, our cell simulations and behavioral experiments indicate that these higher-order low-frequency amplitude modulations, which we refer to as HOLF beats, can be represented within the electrosensory system and drive behavioral responses. We therefore propose that the neural basis of consonance and dissonance perception does not rely on higher-order auditory or cortical processing, but already arises from fundamental mechanisms of neuronal signal transformation at the level of single receptor cells. Furthermore, our findings motivate novel hypotheses about the encoding of fast beat frequencies in higher electrosensory areas and the role of transient frequency modulations in active sensing.

6.1 Spectral Differences Arise from a Single Non-Linearity

Examining the waveforms of superimposed EOD waveforms with different frequency ratios revealed clear differences in the temporal and spectral structure between musical and mistuned

intervals. While all intervals generate a fast first-order beat corresponding to the difference frequency of the underlying signals, mistuned intervals additionally produce a slow amplitude modulation on top of the fast beat (Figure 3, columns 1 and 2). We refer to these slow modulations as higher-order low-frequency (HOLF) beats, and their frequency and amplitude depend on the degree of mistuning as well as the nearest musical interval (Trulla et al. 2018). Musical intervals, in contrast, lack such slow higher-order amplitude modulations entirely.

The power spectra of combined periodic waveforms only produce peaks at the fundamental frequencies and harmonics of the underlying signals. To extract the frequencies of amplitude modulations, the combined waveform must first pass through a non-linearity. We implemented this as a simple cubed threshold function, reminiscent of the synaptic transfer function of auditory inner hair cells (Peterson and Heil 2019). Recent theoretical work demonstrated that this non-linearity is sufficient to extract beat frequencies around the octave in the electrosensory system (Barayeu et al. 2023). This single, biologically plausible non-linearity was also sufficient to generate intermodulation (IM) products in the spectral domain. IM products arise through the interaction of non-overlapping harmonics after non-linear transformations (Anritsu Corporation 2004). In slightly mistuned intervals, harmonics are separated, but remain close to each other. Given Equation 2, their interactions therefore give rise to low-frequency IM products, which manifest as prominent HOLF beats in the temporal domain. With increased mistuning, harmonics will be spaced further apart, such that IM products shift toward higher frequencies (Trulla et al. 2018).

In contrast, most musical intervals generated far fewer IM products than other frequency ratios (Figure 4, A). This is a consequence of the simple integer ratio of the underlying waveforms. The smaller the integers of the ratio, the more harmonics coincide exactly, such that they do not interact to give rise to new frequency components. However, the most dissonant musical intervals, such as the minor second or major seventh, did not differ in IM product count from other frequency ratios. Nonetheless, it appears as if the spacing and amplitudes of IM products is much more ordered in dissonant intervals (Figure 4, C2) than in mistuned intervals (Figure 3 B3, D3, F3).

Together, these findings demonstrate that a single, peripheral non-linearity is sufficient to generate spectral features that distinguish most musical intervals from mistuned intervals. The

same non-linearity that introduces IM components, including those responsible for HOLF beats, is required to extract first-order beats from the signals. Thus, the characteristic features associated with musical intervals do not depend on cochlear or cortical computations, but arise through synaptic transformations already at the single cell level. This insight suggests that low-level, modality-independent cellular mechanisms provide a physiological basis for the spectral differences that emerge between different interval types across species.

6.2 HOLF Beats are Amplified by Neuronal Processing

Given the observed differential temporal and spectral structure between musical and mistuned intervals after non-linear transformation, we wanted to investigate whether these differences are also reflected in the neuronal representation of the respective frequency ratios. To address this, we presented computational models of electroreceptors with stimuli comprising different types of intervals. These simulations revealed that the spiking pattern of electroreceptors encodes both the fast first-order beat and the slower HOLF beat at mistuned intervals (Figure 6, B).

To examine how these frequency components are represented across different timescales of temporal integration, we convolved the spike trains with two Gaussian kernels of different widths. Convolution with a narrow kernel preserves both the fast first-order beat and the HOLF beat. After convolution with a broad kernel, however, only the HOLF beat persists, while the fast first-order beat is effectively filtered out (Figure 6, C). This demonstrates that slow higher-order modulations are robustly represented in the neuronal response even under long integration windows, whereas fast first-order beats are not.

Because convolution acts as a low-pass filter, the standard deviation of a convolved signal reflects the strength of frequency components below the respective cutoff. Comparing the standard deviations of the convolved input signal with those of the convolved spike train can therefore inform us how the signal is transformed through additional cellular non-linearities. For the broad kernel, which indicates the presence of low frequencies, the thresholded input signal shows an increased standard deviation around the unison, octave, and perfect fifth (Figure 8). In contrast, the convolved firing rate shows an increased standard deviation around nearly all musical intervals. Furthermore, the increase in standard deviation appears to be enhanced around the perfect fifth but decreased around the unison and octave as compared to the input

signal itself. This demonstrates that the spike generator of the LIF model does not only retain HOLF beats, but amplifies them beyond the interactions generated through the cubed threshold non-linearity that was applied to the signal.

Overall, these results indicate that HOLF beats are not only preserved across different integration timescales, but their strength is further enhanced through fundamental signal processing mechanisms at the level of individual receptor cells. HOLF beats therefore constitute a prominent feature of mistuned intervals available to the electrosensory system for further processing.

6.3 Encoding Large Difference Frequencies Using HOLF Beats

First-order beats are known to play an important role in the social interactions of weakly electric fish (Henninger et al. 2018; Walz et al. 2013; Zakon and Smith 2002). The mechanisms by which such beats are extracted and encoded in electroreceptors have been described in great detail (Barayeu et al. 2023; Bastian 1981a). Nonetheless, it remains unclear how exactly beats are represented in higher electrosensory areas. Although P-units are capable of following beat frequencies exceeding 300 Hz (Barayeu et al. 2023; Bastian 1981a; Benda et al. 2006; Walz et al. 2014), typical firing rates reported for pyramidal cells in the next processing stage, the electrosensory lateral line lobe (ELL), rarely exceed 50 Hz (Bullock 1969; Chacron 2006). However, experimental data on the representation of high-frequency beats in the ELL is scarce, as most studies to date focussed on difference frequencies below 100 Hz (Bastian 1981b; Krahe et al. 2008). Therefore, it is not obvious how large difference frequencies, which arise predominantly in opposite-sex interactions, could be represented at the level of the ELL.

Our findings offer a possible solution to this problem. As we show, non-linear signal transformations lead to the generation of HOLF beats when fish assume EOD frequencies forming a mistuned interval. These HOLF beats occur at frequencies that fall well within the encoding range of pyramidal cells (Krahe et al. 2008). Hence, we hypothesize that pyramidal cells do not need to encode high frequencies directly. Instead, they could exploit the HOLF beats that emerge naturally at certain frequency ratios through fundamental mechanisms of signal processing at the level of electroreceptors.

The idea that HOLF beats provide an alternative code for large difference frequencies is supported by the analysis of their representations across different integration timescales (Figure 6

and Figure 8). The kernel widths used in this analysis were specifically chosen to resemble integration time scales of P-units ($\sigma = 0.5$ ms) and pyramidal cells ($\sigma = 5$ ms; Berman and Maler 1998), respectively. Hence, the signal obtained through the convolution with a narrow kernel approximates the input pyramidal cells receive from P-units, which contains both the fast first-order beat and the HOLF beat. If pyramidal cells would integrate this input over longer timescales, only the HOLF beat would be extracted. Importantly, our findings further suggest that the strength of HOLF beats is amplified at the level of electroreceptors relative to the cubed and thresholded input signal, making it an even more robust alternative code for large difference frequencies.

Taken together, these findings provide a plausible and empirically testable account of how fast beats are transformed and subsequently represented in higher electrosensory areas. Crucially, the transformations generating HOLF beats do not require dedicated signal transformations, but emerge from fundamental processing mechanisms at the level of individual electroreceptors. In order to test this hypothesis, it is necessary to determine whether pyramidal cells indeed encode HOLF beats in targeted electrophysiological experiments.

6.4 The Behavioral Relevance of HOLF Beats

Given that HOLF beats naturally emerge for a range of frequency ratios and appear to be amplified at early stages of the electrosensory system, we wondered whether fish are actually able to perceive such slow higher-order amplitude modulations. To address this question, we employed a classical behavioral paradigm based on the Jamming Avoidance Response (JAR). The JAR describes a shift in EOD frequency when the fish is presented with another field of similar frequency that produces a slow first-order beat (Heiligenberg et al. 1978; Watanabe and Takeda 1963). We reasoned that if fish are able to perceive HOLF beats, they should elicit a JAR-like frequency adjustment.

Indeed, mistuned intervals evoked JAR-like frequency shifts (Figure 9, Figure 10, A and B). The responses were especially clear for frequency ratios surrounding the perfect fourth and perfect fifth. However, the magnitude of these responses varied substantially across the tested ratios. We could observe that ratios that strongly modulate neuronal responses in our cell simulations also drove the biggest frequency shifts. Ratios that produce weak firing rate modulations, on

the other hand, oftentimes elicited only small or no responses (Figure 10, C). Since the standard deviation of the convolved firing rate captures the strength of low-frequency components in the cell response, this correspondence strongly suggests that the JAR-like frequency adjustments were driven by HOLF beats. Ratios near the major third, however, deviated from this pattern. This could potentially be explained by the range of tested ratios overlapping with other musical intervals, namely the minor third or septimal major third. As a consequence, HOLF beats could become less structured or partially occlude each other. Future follow-up experiments should consequently adjust the tested ratios to each musical interval individually.

In comparison to the unison condition, however, all mistuned ratios generally produced weaker responses (Figure 10). Several factors may contribute to this reduced response magnitude. First, the amplitude of HOLF beats is considerably smaller than that of first-order beats (Figure 3). Previous work from our lab has shown that JAR magnitude scales with beat amplitude (Hladnik 2016). Hence, small HOLF amplitudes could explain small frequency shifts. A second possibility could be different compensation strategies for slow first- and higher-order beats. In a classical JAR, where f_1 lies close to f_0 , fish increase their difference frequency in order to escape the slow first-order beat. At mistuned intervals, however, the HOLF beat can be avoided by either moving away from the musical interval, or converging toward it. Indeed, several individuals robustly converged toward some musical intervals (Figure 11), although this pattern did not appear at the population level. Mating readiness may contribute to the selection of the applied strategy. Anecdotally, we observed that the majority of individuals that converged toward musical intervals appeared gravid, that is, they may have carried mature eggs. This suggests a potential link between reproductive state and the direction of the JAR. Systematically exploring this relationship will be necessary to determine the behavioral contexts in which musical intervals and HOLF beats are relevant to fish.

Apart from the individuals that converged toward musical intervals, response patterns in the low-contrast condition were highly variable across and within fish (Figure 10, A). Contrary to our expectations, fish did not consistently increase or decrease their EOD frequency depending on the position of the stimulus ratio relative to a musical interval. Although such variability was also exhibited in the unison condition, a general trend manifested, which remained lacking for mistuned intervals. One possible explanation for this is that fish failed to compute the sign

of the stimulus frequency relative to the musical intervals. For slow first-order beats, the sign of the frequency shift is computed through the temporal relationship between amplitude and the phase modulation, which are encoded by P-units and T-units, respectively (Carlson and Kawasaki 2007; Heiligenberg et al. 1978). Whether this mechanisms indeed fails at mistuned intervals requires further investigation.

In the high-contrast condition, on the other hand, fish decreased their EOD frequency almost consistently. This may be an effect of the stimulus amplitude substantially exceeding that of the fish’s own electric field. Under these conditions, P-units can lock onto the frequency of the external field. This could lead to a misinterpretation of the own EOD frequency, causing the animal to decrease its frequency in an attempt to return to its baseline. This interpretation, however, is highly speculative, and therefore underscores the need to investigate the mechanisms by which frequency shifts are induced through HOLF beats.

In summary, our behavioral experiments indicate that fish reliably display behavioral responses to HOLF beats, with the magnitude of the response reflecting the strength of the neural representation of the HOLF beat. This provides further support for the idea that fish are able to encode and perceive HOLF beats, while also highlighting the functional and behavioral relevance of slow higher-order amplitude modulations.

6.5 Exploiting Silent Beats During Courtship Season

Although the presented results strongly indicate that HOLF beats hold behavioral relevance, our field observations show that fish frequently tune themselves to musical intervals (Figure 1), which are characterized by the *absence* of HOLF beats. This raises the question of why fish assume exact musical intervals when doing so removes detectable slow amplitude modulations. Both our field data as well as the behavioral experiments (Figure 11) indicate that this behavior is strongly context dependent.

Musical intervals appeared in two out of three of the analyzed dataset. Both of these were collected in Panama during rainy season, whereas the dataset that did not contain precise musical intervals was recorded in Brazil during dry season. From this, it appears as if the formation of simple integer ratios is either limited to locally constrained populations, or may depend on seasonal factors. Notably, rainy season coincides with reproductive activity in many species

of weakly electric fish, such as *Eigenmannia* (Hagedorn and Heiligenberg 1985; Kirschbaum 1975). Combined with reports of fish tuning themselves to musical intervals in laboratory settings (Gaddis 1975), we propose that this behavior is related to reproductive states.

Tuning to musical intervals may provide a means to reduce electrosensory interference during courtship. Our model cell simulations show that longer integration timescales filter out fast beats, whereas HOLF beats at mistuned intervals remain represented. In reproductive contexts, precise electrolocation is crucial. Females have to find a suitable site to deposit their eggs, typically spawning only few at a time, while males must be able to locate eggs accurately to ensure successful fertilization (Hagedorn and Heiligenberg 1985). Since low-frequency amplitude modulations have been shown to interfere with electrolocation (Heiligenberg et al. 1978; Weimar et al. 2025), HOLF beats could be disruptive in these contexts. By tuning themselves to musical intervals, however, fish pairs can minimize the occurrence of detectable low-frequency amplitude modulations, thereby exploiting the spectral properties of simple integer ratios to preserve electrolocation abilities when needed.

To confirm the prevalence of musical intervals during reproductive contexts, future work should aim to elucidate the precise behavioral interactions between fish pairs assuming musical intervals in their natural habitat. Detailed analyses of movement patterns and elicited chirps could reveal whether these ratios coincide with courtship or spawning. Moreover, assessing how stable these intervals remain over time would provide insight into whether fish actively maintain them. Given that the EOD frequency is strongly temperature dependent (Dunlap et al. 2000), fish should drift away from musical intervals as temperature changes. However, if musical intervals are stable over longer periods of time, this would provide strong evidence that fish actively retain simple integer ratios.

6.6 Chirps as an Active Sensing Strategy

Our silent beat hypothesis also offers a novel perspective on the function of transient frequency modulations produced by weakly electric fish. Several types of such modulations have been described, including short chirps (Zupanc et al. 2006) and longer rises (Hagedorn and Heiligenberg 1985; Zakon et al. 2002). Such frequency modulations can be detected by nearby fish as a brief change of the regular AM pattern (Benda et al. 2005; Benda et al. 2006; Vonderschen

and Chacron 2011). As they have been primarily observed during social interactions, they are commonly interpreted as communication signals (Hagedorn and Heiligenberg 1985; Henninger et al. 2018; Zakon et al. 2002; Zupanc et al. 2006). However, this view has been questioned recently, as no consistent social correlate of chirping could be observed in laboratory settings despite large efforts (Oboti et al. 2025).

In light of the silent beat hypothesis, we propose that transient frequency modulations may also serve as a strategy to actively sense the presence of a nearby conspecific. Our findings indicate that higher electrosensory areas cannot encode the conspecific’s EOD when tuned to a musical interval. A transient increase in EOD frequency, however, would shift the ratio from a musical to a mistuned interval. This would briefly generate a HOLF beat, which potentially allows the encoding large difference frequencies in higher sensory areas. Through this mechanism, chirps could provide a means to actively probe the presence or location of a conspecific.

This idea remains speculative though, as we lack in-vivo evidence for how HOLF beats are represented across the electrosensory pathway. Furthermore, it remains unclear whether chirps or other frequency modulations are long enough to generate meaningful HOLF beat patterns. For these reasons, future research should first clarify how musical and mistuned intervals are represented by the nervous system. However, if the silent beat hypothesis proves correct, it provides testable predictions to further elucidate the role of chirps in active sensing.

6.7 Parallels to Consonance and Dissonance Perception

The presented findings on the neural representation and behavioral relevance of HOLF beats in weakly electric fish further provide a novel perspective on the neuronal mechanisms that potentially underlie consonance and dissonance perception. In the electrosensory system, mistuned intervals produce HOLF beats through non-linear interactions between harmonics, whereas musical intervals lack comparable higher-order modulations. Similarly, consonant musical intervals exhibit fewer IM products compared to dissonant intervals (Figure 4 A and B).

We demonstrated that HOLF beats arise from fundamental non-linearities at the level of individual electroreceptors, which share key properties with mammalian inner hair cells, including ribbon synapses and phase-locking to periodic stimulus waveforms (Baker et al. 2013). Importantly, these slow modulations emerge independent of the constraints of the auditory system,

such as the limited frequency resolution of the cochlea. Instead, the spectral differences between musical/consonant and mistuned/dissonant intervals can be explained at the level of single cells. This provides an alternative perspective to classical theories of consonance and dissonance perception that rely on interactions between frequencies within the cochlea, such as Helmholtz’ theory of roughness (Helmholtz 2009).

Furthermore, our modeling results indicate that HOLF beats are not only encoded, but even amplified within the electrosensory system. Furthermore, they could be reliably represented across different timescales of temporal integration. Analogously, studies in the mammalian auditory system have reported strong phase-locking to low-frequency IM products in response to dissonant intervals (Fishman et al. 2001). Hence, both systems appear to be susceptible to low-frequency IM products that differentiate both musical and mistuned intervals, as well as consonant and dissonant intervals.

The occurrence and distribution of IM products at different frequency ratios is closely related to the harmonicity of the given spectrum. Recently, harmonicity has been proposed to play a central role in consonance and dissonance perception (Di Stefano et al. 2022; Harrison and Pearce 2020). However, intervals whose harmonics align frequently naturally produce fewer IM products, yielding a simpler spectral structure. Given this relationship, it is difficult to disentangle whether perceived differences arise from the alignment of harmonics itself or simply from the reduced number of IM components. From a processing perspective, spectra with fewer frequency components should be easier to encode and less computationally demanding for the nervous system, whereas a preference for harmonicity itself would require dedicated processing of the harmonicity of a spectrum. Following the principle of Ockham’s razor, we therefore conclude that the more parsimonious explanation is that intervals with simple integer ratios are preferred due to the reduced number of IM products, thereby allowing a more efficient neural representation than non-musical intervals.

Finally, due to the ambiguous evidence from psychophysical and neurophysiological studies on consonance and dissonance perception, many researchers emphasize the role of cultural experience in consonance preference. While cultural influences most likely indeed influence the *strong preference* for consonant intervals in Western societies, the presence of a bias toward simple integer ratios in weakly electric fish suggests that the basic mechanisms underlying the

perception of consonances and dissonances are evolutionarily conserved and rooted in fundamental principles of neuronal processing. The generation of a larger number of IM products in dissonant intervals relative to consonant intervals arises naturally from non-linearities within individual neurons. This implies that musical intervals are, in a sense, generated through the constraints of neuronal processing, and their perceptual appeal is at least partially rooted in fundamental biophysical mechanisms.

Given the discussed parallels, further research integrating findings about the processing of HOLF beats in weakly electric fish with human neurophysiological and psychophysical studies have a great potential to elucidate the exact neural mechanisms that underlie consonance and dissonance perception. Evaluating these mechanisms will allow the generation of testable hypotheses regarding consonance preference.

6.8 Conclusions and Outlook

The findings of this work demonstrate that a simple, biologically plausible non-linearity implemented at the level of individual receptor cells is sufficient to generate spectral features that distinguish musical from non-musical frequency ratios. This non-linearity also introduces additional frequency components that generate higher-order low-frequency (HOLF) beats in the temporal domain at mistuned intervals. We showed that these slow modulations can be robustly represented across different timescales of temporal integration in the electrosensory system of weakly electric fish. This has profound implications for electrosensory processing. Large difference frequencies naturally occur during opposite-sex interactions, yet how they are represented in higher-order sensory areas remains unknown. Our simulations in a population of electroreceptors strongly suggest that HOLF beats could provide an alternative code for large difference frequencies that can be represented at multiple stages of the electrosensory system.

At the same time, musical intervals are characterized by the complete absence of HOLF beats. This could be exploited by fish to minimize the interference of low-frequency amplitude modulations with electrolocation. This hypothesis is strongly supported by the observation that fish tune themselves to musical intervals under natural conditions and exhibit a JAR-like frequency shift in response to HOLF beats.

More broadly, our results show that the unique spectral structure of musical intervals emerges as a direct consequence of fundamental neuronal signal transformations rather than specialized, modality-specific computations. The parallels between musical and mistuned intervals in fish and consonant and dissonant intervals in humans suggest that the preference for simple integer ratios is an evolutionarily conserved bias that is rooted in the biophysical mechanisms underlying neuronal signal processing. Based on our results, we propose that the relative spectral simplicity of musical intervals leads to a cost-efficient encoding, which is exploited by the nervous system.

Future work should aim to establish these mechanisms on a physiological level. Electrophysiological recordings are needed to clarify if and how HOLF beats are represented at different stages of the electrosensory system and whether they facilitate the encoding of large difference frequencies. However, it is equally important to examine the exact behavioral contexts in which fish assume musical intervals in order to understand their functional and behavioral relevance. If the silent beat hypothesis proves correct, a potential role of transient frequency modulations in active sensing could be further investigated. However, addressing these questions will not only deepen our insights into the electrosensory system, but will also contribute to a broader understanding of how nervous systems extract and represent information from complex periodic signals, and how they might exploit regularities in these signals to achieve efficient neuronal representations. Such insights will furthermore provide novel perspectives on the debate about consonance and dissonance perception.

7 Appendix

7.1 List of Abbreviations

AM	Amplitude Modulation
CLS	Centrolateral Segment
CMS	Centromedial Segment
D1	Dataset 1
D2	Dataset 2
D3	Dataset 3
ELL	Electrosensory Lateral Line Lobe
EOD	Electric Organ Discharge
HOLF	Higher-order low-frequency
IM	Intermodulation
JAR	Jamming Avoidance Response
LIF	Leaky Integrate-and-Fire
LS	Lateral Segment
m2	Minor Second ($f_1 : f_0 = 16 : 15$)
M2	Major Second ($f_1 : f_0 = 9 : 8$)
m3	Minor Third ($f_1 : f_0 = 6 : 5$)
M3	Major Third ($f_1 : f_0 = 5 : 4$)
m6	Minor Sixth ($f_1 : f_0 = 8 : 5$)
M6	Major Sixth ($f_1 : f_0 = 5 : 3$)
m7	Minor Seventh ($f_1 : f_0 = 9 : 5$)
M7	Major Seventh ($f_1 : f_0 = 15 : 8$)
P1	Perfect Unison ($f_1 : f_0 = 1 : 1$)
P4	Perfect Fourth ($f_1 : f_0 = 4 : 3$)
P5	Perfect Fifth ($f_1 : f_0 = 3 : 2$)
P8	Perfect Octave ($f_1 : f_0 = 2 : 1$)
sm7	Subminor Seventh ($f_1 : f_0 = 7 : 4$)
TT	Tritone ($f_1 : f_0 = 7 : 5$)

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